The ABC of Communism: How Soviet Russia Taught Its Women to Read*

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Abstract

How fast can a state teach adults to read, and whom does it teach? In December 1926, 43 per cent of the rural women of Soviet Russia aged 20 to 34 could read. By 1939 literacy in those birth cohorts had risen to 61 per cent, and the census of 1959 still recorded 60. In the thirty years before, the literacy of women of the same ages had scarcely moved. I follow birth cohorts through four censuses, using newly digitised tables of the census of 1926. Relative to older women, the literacy of young women rose by 20 points (standard error 1.7), with gains in every region. A survey of literacy schools in 1924 records how few women they reached in several rural republics, but its regional variation does not explain the later gains with any precision. The censuses establish the scale of the change more securely than they establish the means by which it was achieved. Its attribution to the campaign rests on what would otherwise have happened to learning, selection and the recording of literacy at different ages.

Keywords: adult literacy, education of women, gender gap, state capacity, social mobility, socialism, Soviet Union.
JEL: N34, I25, I28, J16, P36.

*I thank Gerhard Toews and Hosny Zoabi for comments. The 1926 census tables were transcribed from scans held by the Russian State Library. The data, the record of validation of every cell and the code that produces every number in the paper are available at https://github.com/movtish/abc-of-communism. All errors are mine.

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With the transfer of the means of production into common ownership, the single family ceases to be the economic unit of society. Private housekeeping is transformed into a social industry. The care and education of the children becomes a public affair.

Friedrich Engels, *The Origin of the Family, Private Property and the State*

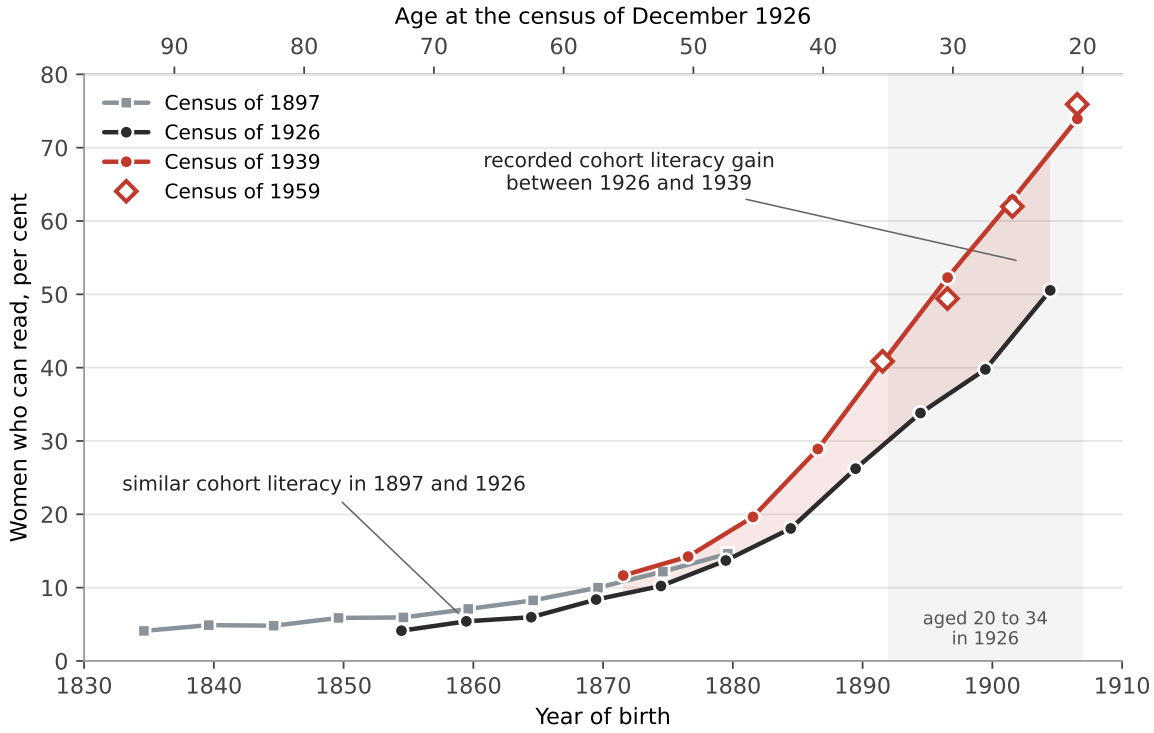
1 Introduction

The history of mass education knows both long periods in which literacy spread slowly, as schooled generations replaced unschooled ones, and short episodes in which a state tried to teach the adults it already had. The Soviet campaign against illiteracy (*likbez*, that is, *likvidatsiya bezgramotnosti*, the liquidation of illiteracy) is a large-scale example of these episodes, and this paper asks how much it actually achieved. In December 1926, nine years after the revolution, 42.6 per cent of the rural women of Russia aged 20 to 34 could read. In January 1939 women of the same birth cohorts were 60.7 per cent literate, and in January 1959, two decades after the campaign, 60.1 per cent. The net gain was almost a third of the cohort's initial illiteracy. How much of it was adult learning, rather than a change in who remained in the cohort, is the question the comparisons below address. Nothing of the kind had happened to their mothers. The rural women who were aged 20 to 34 at the census of 1897 were 10.3 per cent literate in that year and 8.3 per cent literate thirty years later. Figure 1 shows both facts in one picture, and they form the main result of the paper.

The question matters because the education of women is where the largest efforts of states have most often failed. Women still make up the majority of the illiterate adults of the world, and the programmes that have tried to teach them, from national campaigns to the adult classes of development projects, have usually produced small effects or effects that faded (Abadzi, 1994; Ortega and Rodríguez, 2008; Aker et al., 2012; Banerji et al., 2017). The Soviet case records a large increase in the literacy of adult cohorts. The institutional explanations for that increase are the second question of this paper. Among the possible answers one may count a state with the capacity to reach every village, which other states lacked, and incentives for the learners that other programmes did not create, which in the Soviet case took the form of promotion into the posts of the new order (Fitzpatrick, 1979).

That literacy rose under Soviet rule is one of the best known facts of Soviet history, yet the usual evidence for it cannot show what the campaign itself did. The official figures put the literacy of the rural women of the USSR aged 9 to 49 at 35 per cent in 1926 and at 77 per cent in 1939 (Kashin, 1972). An age group of this kind changes its members between

Figure 1: The literacy of rural women by year of birth, as recorded by four censuses



Note: Rural population of the RSFSR. Each point is a five-year age group of the census named, placed at its mean year of birth. The censuses of 1897 and 1926 are those of the 17 regions that made up the RSFSR of 1939, from Table XV of Volume 9 of the census of 1926, which gives the results of 1897 within the borders of 1927. The censuses of 1939 and 1959 are sums over the RSFSR in its borders of those years. The upper axis gives the age of each cohort in December 1926 and the grey band marks the cohort aged 20 to 34. The tinted area is the change in the literacy of the same cohorts between 1926 and 1939. Appendix Figure A8 repeats the picture for the whole population and for men.

two censuses. Girls who had been to the new schools entered it, and their unschooled grandmothers left it. Only a comparison of the same birth cohorts at two dates separates the teaching of adults from the schooling of children. Historians have described the politics and the pedagogy of the campaign (Eklof, 1987; Kenez, 1985; Clark, 2000; Glushchenko, 2015), and regional studies have compared age groups across the two censuses (Zharkov, 2024; Pavlova, 2006), but I know of no measurement on linked cohorts in this literature or in the new economic history of Russia (Zhuravskaya et al., 2024).

To make such a measurement, I link birth cohorts across the censuses of 1897, 1926, 1939 and 1959. The link through 1926 rests on tables of that census which had not been digitised. They give literacy by single year of age and sex for 44 territorial units of the RSFSR, and beside every five years of age they print a subtotal against which each transcribed figure can be checked. The variation I use comes from age and from time. The campaign gave priority to younger adults, and I compare their gains with those of older cohorts. The comparison

is informative to the extent that differences in ordinary learning, mortality, migration and reporting would not have produced the same age pattern without the campaign. The cohorts of the same ages a generation earlier show what happened when no campaign reached women at all. Section 4.5 examines what could make this comparison misleading, from selective mortality and migration to an exaggerated count in the census of 1939.

The estimates confirm the size of the change. In the 18 units where cohort age windows can be linked, the literacy of rural women aged 18 to 32 in 1926 rose from 41 to 65 per cent by 1939. Relative to women above the age limit, the rise is 20.4 points (standard error 1.7), or 37 per cent of those who could not read, and it lies between 18.0 and 21.5 points in every sample of units. Literacy rose in every region of the republic. In the RSFSR as a whole about 2.6 million women aged 20 to 49 in 1926 learned to read between the two censuses, and 1.1 million men.[1] Women made up 71 per cent of those who learned. The census of 1959 records sustained high literacy among the surviving cohorts, although subsequent learning, selection and territorial changes prevent it from dating all of that learning to the campaign.

Why had the literacy of adult women stood still for thirty years, and why did it move then? The Bolsheviks had an answer before they had a campaign, and they took it from Engels. In a society whose economic unit is the household, a wife's labour is a private service to her family, and her emancipation requires that the functions of the household, among them the upbringing and education of children, become a public affair (Engels, 1884). *The ABC of Communism*, the commentary on the party programme of 1919 from which this paper takes its title, drew the conclusion that the right to educate belongs to society before it belongs to parents, and called for all who could read to be mobilised to teach all who could not (Bukharin and Preobrazhensky, 1919). For a decade, however, the Soviet state offered classes and left attendance to the household, as the schools of the empire had done, and the women stayed at home. The large recorded gains belong to the interval that includes the registration of illiterate villagers, the enforcement of attendance and collectivisation. The household account gives these changes a common logic, though two censuses cannot establish either a break in 1929 or the separate effect of each change.

The second half of the paper asks what can be learned about this institutional interpretation. I bring together the age profile, the earlier experience of men, regional urbanisation and a newly transcribed survey of literacy schools in 1924. The survey measures institutions, teachers and pupils by sex in fifteen territories. Fourteen can be linked to a validated cohort in the census data. It gives the argument a direct institutional comparison, but a small and incomplete one. The later gains cannot be attributed separately to enforcement, household

[1]The change in the literacy of each five-year group of the whole population between 1926 and 1939, multiplied by its number in 1939 and summed over the groups aged 20 to 49 in 1926.

decisions or private returns on this evidence.

The paper contributes to four literatures. Historians of the campaign (Eklof, 1987; Clark, 2000) describe its organisation and judge its results from the reports of the agencies that ran it, which counted enrolment and had no way of counting readers. This paper measures the proportion of a cohort recorded as literate at successive censuses. The economics of adult literacy finds small or no effects of programmes in Venezuela, Niger and India (Ortega and Rodríguez, 2008; Aker et al., 2012; Banerji et al., 2017), and the Soviet case suggests why. Those programmes offered classes and left attendance to the household, as the Soviet state itself had done in the 1920s with the same result. Unlike the Cuban campaign of 1961, whose effects cannot be separated from the health reforms of the same years (Mellace and Spruk, 2026), the Soviet case offers the age limit and the earlier generation as counterfactuals. In addition, the result bears on the origins of gender gaps in human capital, which a large literature traces to persistent culture (Alesina et al., 2013). Finally, the paper extends the study of nation-building through education (Weber, 1976; Bandiera et al., 2019; Alesina et al., 2021; Blanc and Kubo, 2023) from children to adults. France made one nation out of peasants who spoke Breton or Occitan by sending their children to the schools of the state, while the Soviet state taught the adults themselves and reached the women whom the schools of the past had reached least. It thereby adds a measured case to the study of state capacity and of the legacies of socialism for women (Besley and Persson, 2009; Charnysh, 2019, 2024; Campa and Serafinelli, 2019; Lippmann et al., 2020).

The paper proceeds as follows. Section 2 describes the doctrine and the campaign. Section 3 presents the data and the design and Section 4 the results. Section 5 asks why the campaign worked, Section 6 discusses what the result implies and what it does not, and Section 7 concludes. The appendix contains the validation of the data, the model and further estimates.

2 Doctrine and Campaign

2.1 From Engels to the ABC of Communism

The Soviet view of women and the family had a clearly identifiable source. *The Origin of the Family, Private Property and the State*, first published in 1884, was the text from which Russian Marxists took their view of the family, and its argument consists of two parts. The first part is a diagnosis. Where the single family is the economic unit of society, the management of the household is "a private service" and the wife "the head servant, excluded from all participation in social production". The second part is a remedy. Her emancipation requires her return to public production, and this in turn requires that the family cease

to be the economic unit, so that housekeeping becomes a social industry and "the care and education of the children becomes a public affair" (Engels, 1884, ch. 2). Translated into the language of economics, the diagnosis says that the skills of a woman earn nothing her household can see, so that the household will not pay to improve them. The remedy, accordingly, says that the decision must pass to someone who values those skills.

The Bolsheviks turned this remedy into a programme. *The ABC of Communism*, written by Bukharin and Preobrazhensky in 1919 to explain the new party programme to its members, states that "the first and fundamental right to educate children belongs to society", and it lists among the tasks of the Soviet power schools for illiterate adults and "the mobilisation of all the literate to teach all the illiterate" (Bukharin and Preobrazhensky, 1919, §§79 and 84, my translation). In the same year Lenin wrote that the law had made woman equal, while housework still kept her "a domestic slave" (Lenin, 1919). The decree of 26 December 1919 obliged every citizen between the ages of 8 and 50 to learn to read and write, in Russian or in the native language, "in order to give the whole population of the Republic the opportunity of conscious participation in the political life of the country" (Sovet Narodnykh Komissarov RSFSR, 1919). The women's departments of the party carried this argument to the village (Wood, 1997; Goldman, 1993), and the campaign added one of its own. A poster of 1923 shows a schoolgirl at her homework telling her mother that she could have helped, had she been able to read (Kruglikova, 1923). The doctrine, in other words, was in place long before the means to carry it out.

2.2 Two instruments

The decree of 1919 was carried out under two different arrangements, which may be called two instruments. Throughout the 1920s the literacy points (*likpunkty*), the Extraordinary Commission for the Liquidation of Illiteracy and the society Down with Illiteracy offered classes with limited funds, and in the village attendance was not enforced. The adults they reached were mostly men. The army taught its conscripts, the unions taught workers, and the village literacy point was attended by young men who expected to leave for the town or the army, and by few married women (Clark, 2000; Eklof, 1987; Kenez, 1985). In this respect the first Soviet decade continued the arrangement of the late empire, whose *zemstvo* and parish schools had taught those whom a household released to them.

The instrument changed at the end of the decade, with the cultural campaign launched by the Komsomol in 1928, the collectivisation of agriculture from 1929 and the decrees on universal primary schooling of 1930. The illiterate of each village were registered, classes were attached to the collective farm, cultural brigades went from house to house, and attendance was enforced (Fitzpatrick, 1979; Eklof, 1987). The change is visible in enrolment. The

schools for the illiterate, which had enrolled about one and a half million persons a year in the USSR between 1926 and 1930, enrolled five to six million a year between 1930 and 1932 and three to five million a year until 1936. The schools first took those aged 16 to 35 and then the older groups up to 50, and a decision of 1936 set the liquidation of illiteracy among those under 50 as the task of that year and the next (Bol'shaya sovetskaya entsiklopediya, 1938). The same years opened the posts of the new order to literate workers and peasants. Workers' faculties (*rabfaki*), which prepared adults for higher education, had existed since 1919, and under the First Five-Year Plan literate workers and peasants were promoted on a large scale into management, the party and the offices of the collective farms (Fitzpatrick, 1979). The campaign thus coincided with a wide opening of what later came to be called social lifts.

The timing of the censuses fits these two arrangements closely. The census of December 1926 was taken at the end of the first decade and the census of January 1939 at the end of the second. Thus the cohorts that were adults in 1926 show what the second instrument did, while the cohorts that were adults in 1897 show what happened to adult women as long as attendance was left to the household, under the empire and in the first Soviet years alike.

3 Data and Design

3.1 Four censuses

The four censuses used in the paper record literacy in comparable ways. The censuses of 1897, 1926 and 1939 define it in the same way, as the ability to read in any language (Tsentral'noe Statisticheskoe Upravlenie SSSR, 1928; Zhiromskaya et al., 1996). The census of 1959 recorded schooling and, for those without schooling, the ability to read.

The core of the new data comes from the regional volumes of the census of 1926. For every *guberniya*, *okrug* and autonomous unit they print a table of the population by single year of age, sex and literacy, separately for town and village (Table IX). These tables had not been digitised before. I transcribed them from the scans of the Russian State Library for 44 units, 90,642 cells in all. Every five-year figure appears in them twice, as a printed subtotal and as the sum of five single years transcribed separately, and a figure enters the analysis only if the two agree or if another identity of the table decides between them. Of the 8,840 five-year groups between the ages of 10 and 69, 91.8 per cent pass the first test and 6.3 per cent are dropped (Appendix A).

Two further sources complete the data. Volume 9 of the same census recomputed the results of 1897 to the borders of 1927 for the nineteen regions of the RSFSR, by sex and five-year age group (Table XV). The tables of 1939 by region, age, sex and rural or urban

residence, and the national and regional tables of 1959, were declassified from the Russian State Archive of the Economy and digitised by Demoscope Weekly.

Two sources describe the campaign itself. The survey of educational institutions on 1 January 1924 counts literacy points and schools, their teachers and their pupils, separately by sex and for town and village (Central Statistical Administration, 1926). I transcribe thirty rows for fifteen territories. The economic plan for 1931 gives the ages to which instruction was addressed and records the difference between enrolment and completion (Gosplan SSSR, 1931). Neither source counts the adults whom the census later found able to read. Their purpose here is to compare the census evidence with the institutions and priorities of the campaign.

The regional tables also require a distinction that their headings can obscure. The age table of 1939 includes Moscow and Leningrad within their oblasts; the illiteracy table reports the cities separately. I subtract each city's population from its oblast before joining the tables, and add the city once when forming larger regions. This changes urban and total-population results, including the regional comparison of urbanisation, while leaving the rural cells unchanged.

3.2 Linked cohorts

Linking cohorts across censuses requires matching their dates and their territories. The interval from December 1926 to January 1939 is twelve years and one month. I approximate it by an age shift of twelve years; respondents with birthdays in the intervening month can be thirteen years older. From the validated single years of 1926 I build cohorts aged 18 to 22, 23 to 27 and so on up to 53 to 57, and I compare each with the five-year group of 1939 that is twelve years older. In 18 cases a unit of 1926 can be linked to a successor of 1939 and has validated rural pages, which gives 218 cells by unit, cohort and sex. This is the main sample. It is rural because four fifths of the population lived in the countryside, and because it was in the village that the campaign met the household.

A second sample covers the whole of the RSFSR. For the 17 regions of Table XV that formed the RSFSR of 1939, a five-year group of 1926 is matched to three fifths of one five-year group of 1939 and two fifths of the next, and in the same way to the census of 1959. A third sample links the five-year groups of 1897 to the groups thirty years older in 1926, in 19 regions.[2]

[2]The census of 1897 was taken twenty-nine years and ten months before that of 1926. Figure 1 uses the regions of the second sample and plots every census with its own age groups, so that nothing in it is interpolated.

3.3 Design

The change in the literacy of a cohort combines learning with changes in its membership and its enumeration. An age comparison can distinguish these only under an assumption about the change that younger and older adults would have experienced without the campaign. The campaign was addressed to those under 50. The cohorts aged 48 to 57 in 1926 had passed that age when its mass phase began, so the change in their literacy measures selection, together with whatever was taught above the limit. With Δy_{ucs} the change in the literacy of cohort c of sex s in unit u, I estimate

$$\Delta y_{ucs} \;=\; \mu_u \;+\; \sum_{k\in\{Y,M\}} \left(\beta_k D_c^k \;+\; \delta_k D_c^k F_s\right) \;+\; \phi F_s \;+\; \varepsilon_{ucs}, \tag{1}$$

where D_c^Y indicates the cohorts aged 18 to 32, D_c^M those aged 33 to 47, F_s indicates women and μ_u are unit effects. The younger-minus-older contrast is β_Y for men and $\beta_Y + \delta_Y$ for women; δ_Y is the difference between them. If older adults were unaffected and counterfactual age differences were absent, the first two contrasts would give the effects on younger men and women. Since men also received instruction, the triple difference would still measure a difference between effects, rather than the effect on women alone. Observations are weighted by the population of the cohort in 1926, and standard errors are clustered by the shared 1939 source territory for the unit sample, and by region for regional comparisons. I estimate the same equation for the period before the campaign, comparing the cohorts aged 10 to 24 in 1897 with those aged 30 to 44.

A change in percentage points depends on the room left below 100 per cent, and this room differed between the sexes. A second outcome is therefore the net literacy gain relative to initial illiteracy, $t = \Delta y/(100-y^0)$, where y^0 is the literacy of the cohort at the earlier census, weighted by the number of its illiterate. The female difference-in-differences identifies a campaign effect only if, absent the campaign, younger and older women would have experienced the same conditional change in recorded literacy. This restriction concerns all age-specific learning, selection, reporting and territorial composition, not mortality alone. The triple difference requires a different restriction: the counterfactual younger-minus-older contrast must be equal between the sexes. Because men were also exposed, this contrast identifies a differential effect by sex rather than the total effect on women. Section 4.5 reports accounting and specification sensitivities; they cannot verify these identifying restrictions. Since the main sample has only 18 units, the estimates are accompanied by leave-one-out ranges, a permutation test and a wild cluster bootstrap (Cameron et al., 2008). The earlier specification was replicated in Python and Stata; the present data correction and supplemental

inference are implemented in Python.

4 Results

4.1 The main picture

Figure 1 plots the share of rural women who could read against their year of birth, as each of the four censuses recorded it, so that the vertical distance between two lines is the change in the literacy of a cohort.

The censuses of 1897 and 1926 trace one and the same line. The women born between 1850 and 1880, who were adults in 1897, were as literate in 1926 as they had been thirty years before, to within 2.3 points in every five-year group. Literacy rose from one cohort to the next, as the school reached more girls, yet it did not rise within a cohort. This line serves as the counterfactual for what follows. In rural Russia a woman who had not learned to read as a child did not learn later.

The census of 1939 departs from that line, by 19 points for the women aged 20 to 24 in 1926, by 7 for those aged 40 to 44 and by 3 for those aged 50 to 54, who had passed the age limit (Appendix Table A8). The points of the census of 1959 lie on the line of 1939. Taken together, the cohort aged 20 to 34 in 1926 went from 42.6 to 60.7 per cent, while the cohort that had been of the same ages in 1897 went from 10.3 to 8.3.

4.2 The size of the effect

The estimates for the units in which cohorts are matched exactly confirm what the figure shows (Table 1). Of the rural women aged 18 to 32 in 1926, 41 per cent could read in that year and 65 per cent in 1939. Among the women aged 48 to 57 literacy rose from 8 to 13 per cent. Within units, the difference between the two groups of cohorts is 20.4 points (standard error 1.7). Of the younger women who could not read in 1926, 41 per cent could read in 1939, against 6 per cent of the older women, and the difference within units is 37.2 points (2.1).

The men of the same cohorts present a different picture. Their literacy rose from 82 to 93 per cent. An illiterate man was more likely to be taught than an illiterate woman, 60 per cent against 41, but the campaign found few illiterate men, and in points the gain of women exceeds that of men by 14.4 (2.1). Within the cohort aged 23 to 27 the gap between the sexes fell from 45 to 30 points, while among those aged 43 and over it did not move.

Panel B of the table covers the thirty years before the campaign. The women aged 10 to 24 in 1897 were 14 per cent literate in that year and 14 per cent in 1926, and the older

Table 1: The same rural cohorts at two censuses, during and before the campaign

	Literate, per cent		Change,	Illiterate who
	earlier census	later census	points	learned, per cent
A. The campaign, 1926 to 1939 (18 units)				
Women aged 18 to 32 in 1926	41	65	24.4	41
Women aged 48 to 57, above the age limit	8	13	5.2	6
Men aged 18 to 32 in 1926	82	93	11.1	60
Men aged 48 to 57, above the age limit	46	52	6.4	12
Women, younger minus older cohorts			20.4 (1.7)	37.2 (2.1)
Men, younger minus older cohorts			6.0 (1.3)	49.6 (2.5)
Women minus men			14.4 (2.1)	−12.5 (2.7)
B. Before the campaign, 1897 to 1926 (19 regions)				
Women aged 10 to 24 in 1897	14	14	0.1	0
Women aged 30 to 44 in 1897	7	5	−1.8	−2
Men aged 10 to 24 in 1897	42	59	17.2	30
Men aged 30 to 44 in 1897	32	31	−1.1	−2
Women, younger minus older cohorts			2.0 (0.4)	2.2 (0.5)
Men, younger minus older cohorts			18.3 (1.4)	31.3 (3.0)
Women minus men			−16.3 (1.5)	−29.1 (2.8)

Note: Rural population. Literacy rates are means of the cohorts weighted by their population at the earlier census. The last column is the change in literacy divided by the share illiterate at the earlier census, weighted by the number of illiterate. The rows of differences are the coefficients $\beta_Y + \delta_Y$, β_Y and δ_Y of equation (1), with standard errors clustered by the shared later-census territory or region in parentheses. Panel A links cohorts built from single years of age in 1926 to the five-year groups of 1939 twelve years older, and its regressions include the cohorts aged 33 to 47 with their own coefficients. Panel B links the five-year groups of 1897 to the groups thirty years older in 1926 and leaves out the cohort aged 25 to 29.

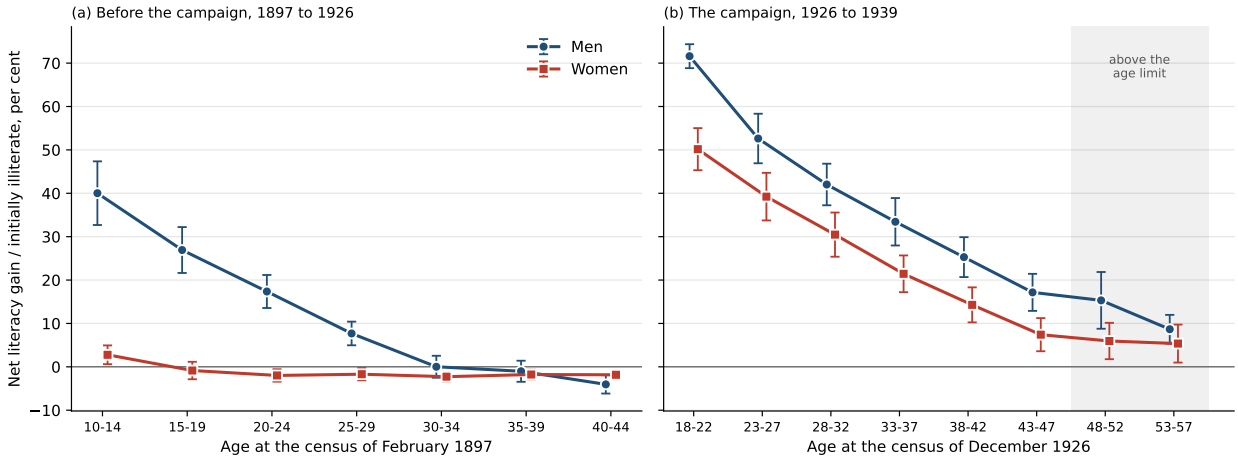
women 7 and 5 per cent. The difference between them, 2.0 points, belongs to the girls aged 10 to 14, who were still of school age. The literacy of the men aged 10 to 24, in contrast, rose from 42 to 59 per cent. Thus men learned as adults under the old arrangement, in the army, at work and in the town, while women did not.

The number of independent later-census territories is smaller than the number of units in the transcription. With inference clustered on those 17 territories, the CR2 interval for the female contrast is [16.68, 24.03] points, and that for the triple difference is [9.74, 19.02]. These intervals allow for the limited number of clusters; they do not remove bias from the cohort comparison itself.

4.3 Age and sex, before and during the campaign

The age profile of learning provides a further check on the argument (Figure 2). The figure shows the share of the illiterate who learned, by age at the earlier census. Before the campaign the share of women is nil at every age from 15 to 44, while that of men falls from

Figure 2: The share of the illiterate who learned to read, by age, before and during the campaign



Note: Change in the literacy of the cohort divided by the share illiterate at the earlier census, rural population, with 95 per cent confidence intervals from standard errors clustered by region or unit. Panel (a): 19 regions, 1897 to 1926. Panel (b): 18 units, 1926 to 1939. The shaded cohorts were above the age limit of the campaign.

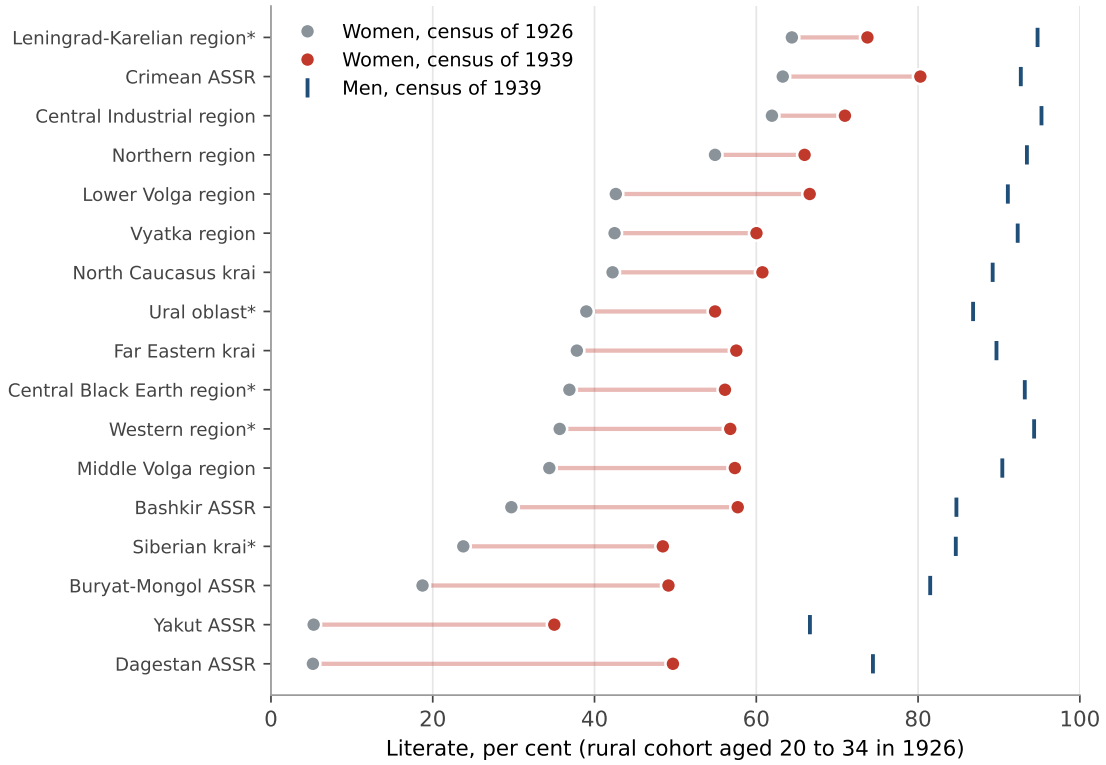
27 per cent at ages 15 to 19 to 8 per cent at 25 to 29 and to nothing above 30. During the campaign the share of women is 50 per cent at ages 18 to 22, 39 at 23 to 27 and 30 at 28 to 32. It falls by 1.77 points with each year of age, to 7 per cent at 43 to 47 and 5 to 6 per cent above the limit.

The share of men lies above that of women at every age, so the campaign did not favour women. It reached the illiterate of both sexes, and men kept in addition the routes to literacy that they had always had (Section 5). What distinguishes the two periods is the appearance of women among the taught.

4.4 Every region

The rise in the literacy of women was not confined to a few regions (Figure 3). The figure follows the cohort aged 20 to 34 in 1926 in each of the 17 regions of the RSFSR. The literacy of rural women rose in all of them, by between 9 points in the Central Industrial region and 45 points in the Dagestan ASSR, and between 24 and 47 per cent of the illiterate learned to read. The gain in points was larger where fewer women could read in 1926, while the share of the illiterate who learned varied far less, a pattern to which Section 5 returns. The smallest gains are those of the regions around Moscow and Leningrad, where most women could already read, and the largest those of four autonomous republics. It is worth stressing, however, that the campaign was no nationality policy in disguise. In the Russian *gubernii* and *okrugi* of the main sample the effect on women is 20.7 points (2.0), and in its autonomous republics and oblasts 20.5 (3.4).

Figure 3: The cohort aged 20 to 34 in 1926 at the censuses of 1926 and 1939, by region



Note: Rural population of the 17 regions that made up the RSFSR of 1939, ordered by the literacy of women in 1926. The figures of 1926 are those of Table XV of Volume 9 and the figures of 1939 those of the oblasts of 1939 aggregated to regions, each five-year group of 1926 being matched to three fifths of one group of 1939 and two fifths of the next. The vertical marks give the literacy of the men of the same cohort in 1939. An asterisk marks a region whose borders of 1939 differ from those of 1926.

The age limit was a changing administrative priority, rather than a single threshold maintained throughout the campaign. The plan for 1931 addressed those aged 16 to 50 in towns and 16 to 40 in the countryside, with some extension above 40 in collectivised areas. The cohort aged 33 to 37 in 1926 straddled the rural limit by 1931. It belongs to the intermediate group in equation (1). Omitting it leaves the younger group at 18 to 32 and changes the female contrast from 20.354 to 20.312 points. The triple difference changes from 14.379 to 14.377. The comparison is therefore little affected by that intermediate cohort. It does not follow that no older adult was taught.

4.5 Is the rise real?

The size of the effect raises the question of whether it is genuine, and several checks address it. Table 2 repeats the estimate in samples that differ in weighting, in the treatment of cohorts whose size changed between the censuses and in the units excluded. The effect on

Table 2: The effect of the campaign on the literacy of women in other samples

Sample or specification	Women, younger minus older cohorts: points	Women, younger minus older cohorts: illiterate who learned, per cent	Women minus men, points	Units; cells
Baseline	20.4 (1.7)	37.2 (2.1)	14.4 (2.1)	18; 218
Unweighted	20.0 (1.5)	35.6 (2.5)	14.2 (2.1)	18; 218
Controlling for the change in the size of the cohort	20.1 (1.7)	36.8 (2.2)	13.2 (2.1)	18; 218
Cohorts whose size changed little	18.0 (1.8)	34.4 (2.5)	10.2 (1.9)	12; 124
Russian *gubernii* and *okrugi* only	20.7 (2.0)	37.4 (2.5)	14.7 (2.0)	12; 138
Autonomous republics and oblasts only	20.5 (3.4)	37.3 (4.4)	15.6 (5.0)	6; 80
Without the Tatar and Bashkir republics	21.5 (1.8)	38.4 (2.3)	16.2 (1.9)	16; 194
Without the Ural *okrugi*	20.4 (1.7)	37.5 (2.1)	14.5 (2.2)	15; 178
Without the units with low-quality pages	20.0 (1.9)	37.4 (2.4)	14.7 (2.4)	15; 192
Whole population of the units	18.5 (1.0)	36.8 (1.4)	13.2 (1.4)	25; 352
Regions covering the whole rural RSFSR	14.6 (2.0)	27.2 (1.5)	10.0 (1.6)	17; 238
Regions with unchanged borders	13.6 (2.6)	27.2 (2.2)	8.5 (2.0)	12; 168

Note: Coefficients of equation (1) for 1926 to 1939, rural population unless stated, with standard errors clustered by the shared later-census territory or region in parentheses. The first two columns give $\beta_Y + \delta_Y$, the change among women aged 18 to 32 in 1926 relative to women aged 48 to 57, in points of literacy and as a share of the illiterate, and the third gives δ_Y. The cohorts whose size changed little are those with between 0.6 and 1.15 times their number of 1926 in 1939. In the last two rows the units are regions, the younger cohorts are aged 20 to 34 and the older cohort is aged 50 to 54.

women lies between 18.0 and 21.5 points, and between 34 and 38 per cent of the illiterate. In the 17 regions that cover the whole rural RSFSR it is 14.6 points (2.0) and 27 per cent. It should be noted that the cohorts of that sample are two years older and its comparison cohort, aged 50 to 54, is younger. With a separate effect for every cohort in place of the two indicators, the triple difference of the main sample is 14.5 (2.1). It lies between 13.3 and 15.8 when each unit in turn is excluded, and a wild cluster bootstrap gives a Monte Carlo p-value of 0.001 (Appendix E).

The census of 1939 deserves particular attention, since it followed the suppressed census of 1937 and was taken under pressure. The overcount of its population, which Zhiromskaya (2001) puts at 1.7 per cent, would raise a literacy rate by less than two points even if every person added had been entered as literate. In addition, a uniform proportional reporting error alone would not explain a strongly declining age profile, but age-specific error remains possible. The recorded normalised gain falls from 50 per cent among the youngest women to 5 per cent among the oldest. An additional consistency check is the census of 1959. In the whole population of the RSFSR, which migration between village and town does not affect, the literacy of the women aged 20 to 34 in 1926 was 51.2 per cent in that year, 69.7 in 1939 and 67.6 in 1959. Measured on the census of 1959 alone, 34 per cent of those who were illiterate in 1926 had become literate, against 38 per cent measured in 1939. In the rural population the two figures are 31 and 32 per cent (Appendix Table A7). The 2.0 point difference between the two later censuses is a net outcome of possible reporting error in

1939, subsequent learning, loss of skills, selective survival, migration and territorial changes. Without independent restrictions on these components it does not bound the reporting error of 1939. The 1959 figures support persistence of high cohort literacy, while the timing of acquisition remains uncertain.

The regional tables of 1959 make this caution more concrete. In five republics, the literacy of the female cohorts aged 30 to 44 in 1939 rises by 3.78 points in the countryside but changes by -8.10 points in towns by 1959. These age bands can be followed twenty years without interpolation, though they do not coincide exactly with the younger group of 1926. The urban decline need not mean that women forgot how to read. In the Tatar republic the number of women in the urban cohorts is larger in 1959 than in 1939, so the comparison necessarily includes a change in who belonged to them. The national persistence of high literacy and the regional movement of recorded rates are compatible with one another (Appendix H).

Selective mortality and migration are the last concern. The years between the censuses include collectivisation, the famine of 1932 to 1933 and the departure of millions from the village. The count of the later cohort is not a survival count when its territory changes. In particular, two Ural districts are compared with the same larger oblast. I therefore do not use the pooled ratio of their populations as an upper bound on selective attrition. Even for matched territories a bound based on net attrition requires restrictions on compensating inflows. The main female estimate remains positive when the Ural districts are excluded and in restricted samples, but this robustness does not rule out all age-specific selection. The entire-population results remove migration between village and town as an accounting change, while retaining cross-border migration and changes in the national territory.

5 Institutional Explanations and Their Limits

5.1 Three explanations

Three explanations of the observed cohort changes generate overlapping predictions and organise the discussion; the available variation does not distinguish all of them. The first is the capacity of the state, on which account the gains should appear with the means of enforcement. The second is incentives. If literacy opened the posts of the new order, the return to a woman's literacy rose for her and for her household, and learning should be concentrated among the young, who had most to gain, and in the regions where the new economy offered most. The third is the transfer of the decision from the household to the state. It predicts that the reach of the state depends on the age of the learner and hardly on local opportunities, and that men learned at the rate of that reach plus the rate at which

they had always learned.

5.2 Two conditions

The argument of Engels can be written as two inequalities, which Appendix B develops into a full model and which formalise the third explanation. A household teaches an illiterate member of sex s and age a if what literacy will bring the household exceeds its cost in time and effort,

$$r_s \, A(T-a) \; + \; \mathbf{1}\{s = f\} \, \alpha\gamma \; \geq \; c, \tag{2}$$

where r_s is the yearly return to literacy that the household receives, $A(T-a)$ the discounted length of the remaining working life, γ the value of a literate mother to her children and $\alpha < 1$ the weight that the household gives to it. Low private returns to women's literacy are one possible parameterisation, not a measured consequence of unpaid household labour. Little adult learning follows only if private and intergenerational benefits jointly fall below costs for most households. The net literacy changes observed in census cohorts do not separately identify these returns, costs or household bargaining weights.

The Soviet state valued the same literacy differently. It teaches an illiterate adult if

$$r_s \, A(T-a) \; + \; \mathbf{1}\{s = f\} \, \gamma \; + \; \pi \; \geq \; c_S(a), \tag{3}$$

where π is the political return to a citizen who can read, that is, the return named in the preamble of the decree, and $c_S(a)$ is the cost of teaching an adult of age a, which rises with age. The term π does not depend on sex. If it is large relative to the other terms, as the scale of the campaign suggests, the state teaches illiterate women and men at rates of the same order, and it teaches more of the young than of the old. Since most of those whom the household left illiterate are women, most of those whom the state teaches are women as well.

The result does not require γ, the return through children on which economists usually rest the case for teaching mothers (Galor, 2011). The political return named in the decree is sufficient, although the campaign used the other argument as well, and the data cannot tell the two apart. The decree of 1919 already contained π, yet as long as it was carried out by offering classes, the choice of who attended remained with the household and condition (2) governed. Condition (3) began to govern only when the campaign, so to speak, "went to the door".

Taken together, the two conditions give three predictions. Before the campaign reaches the household, the literacy of adult women does not change and that of young men rises. Under the additional restrictions of negligible household learning among women, common

state reach and comparable populations, the share of illiterate women who learn equals $e(a)$. Its age profile depends on both benefits and costs; independence from private returns is an assumption to be investigated rather than a consequence of the observed campaign scale. The share of men equals that reach plus the household's own rate H_m for the men whom the state does not reach,

$$P_m(a) \;=\; e(a) \;+\; \Big(1 - e(a)\Big)\, H_m(a). \tag{4}$$

5.3 Evidence

The timing of the gains supports the first explanation in a qualified form. The decree, the literacy points and the promotion of literate peasants all existed in the 1920s, yet the women aged 38 to 52 at the decree were no more literate in 1926 than in 1897 (Figure 2). The gains came with the instrument of the 1930s, which the collective farm made possible by giving the state an organisation in every village. This timing is consistent with an expanded capacity to organise instruction. The data do not establish that capacity was necessary, or that it acted through a transfer of the attendance decision rather than through other changes of the period. The survey of 1924 offers an earlier measure of the institutions of instruction. It does not measure their expansion or enforcement in the years of the largest gains.

The survey gives a striking picture of how unevenly those early institutions reached women. In the rural Tatar republic, all 81 registered literacy points and schools were surveyed, and women were 12 per cent of their pupils. In the Mari territory they were 67 per cent, but only five of 23 institutions were surveyed. The second fraction cannot be read as broader access without knowing whom the missing institutions taught. Nor can a count of pupils on one date be read as a count of graduates over a decade.

I relate the subsequent female-minus-male gain in the youngest validated cohort to the female shares of pupils and teachers, controlling for the initial literacy gap. I also compare the numbers of institutions and teachers, allowing for the size of the cohort. Across twenty-four combinations of measures, weights and coverage restrictions, every CR2 interval includes zero. Some bootstrap tests give smaller probabilities, but the result does not survive the changes of sample and weighting. The early network does not predict the later relative gain with useful precision in this sample. This leaves open both the effect of instruction and the importance of the network built after 1924. The full transcriptions and estimates are reported in Appendix H.

The share of men permits an illustrative accounting exercise for the third explanation. It is not an independent test of the mechanism: the historical male probability is measured over about thirty years, compared with twelve years during the campaign, and its transport to men whom the state would not reach requires additional assumptions. If the reach of the

state among women, 0.41, applied to men as well, and the men whom it did not reach learned at the rate of men of the same ages before the campaign, 0.18, equation (4) predicts that a share of 0.52 of illiterate men learned. The observed share is 0.60. Under this transport assumption, the arithmetic assigns 11 of the 19 points to the historical male component. This allocation is not identified: it changes with the exposure horizon and selection into the group not reached by the state (Appendix E).

The age profile alone cannot distinguish incentives from the organisation of instruction. Private benefits need not consist only of a new career, and the opportunity cost of attendance can vary with age and family composition. If the gains had come from the prospect of advancement, learning would have been concentrated among the young women for whom a new career was plausible. Yet 21.4 per cent of the illiterate women aged 33 to 37 in 1926 learned, and 14.3 per cent of those aged 38 to 42, women who were in their forties by the end of the campaign and to whom literacy promised little of the kind.

The regions provide the last test. Across the 17 regions of the RSFSR, the share of the cohort aged 20 to 34 in 1926 that lived in towns rose by between 5 and 26 points between the censuses, which measures how far the opportunities of the new economy reached each region. In the rural population the share of illiterate women who learned is lower where more of the cohort moved to the towns, by 1.8 points per standard deviation (standard error 1.4), as it would be if those who learned were among those who left. In the whole population, which the move to the town does not affect, the correlation is 0.12 across all regions, a figure that rests on the Leningrad region, which drew young migrants from other regions, and -0.12 in the 12 regions whose borders did not change (Appendix Figure A7). These associations do not identify whether learning caused migration, migration changed learning opportunities, or both responded to regional change. A weak association in a small cross-section is not evidence of equivalence to zero. The corrected territorial aggregation and additional small-sample inference are reported in the methodological supplement.

Taken together, the evidence is consistent with a combination of expanded instruction, enforcement and changing private returns. It does not rank these explanations causally. A stronger test would require measures of local exposure, implementation or attendance that precede the measured outcomes, together with a defensible comparison design. The household model organises hypotheses about these channels; it does not establish which channel generated the observed cohort changes.

6 Discussion

The effect is far outside the range of the adult literacy programmes that economists have evaluated, and the features that distinguish the Soviet campaign are fairly easy to name. Among them one may count enforced attendance, with its cost borne by an institution other than the learner, namely the collective farm and the school. Another feature was duration, since the campaign ran for a decade, while most modern programmes last a season. Finally, most of its learners were young adults. None of these features is peculiar to the Soviet case. What is peculiar to it is a state that applied all of them to every village of a country, and that had explained in advance why it would do so. Compulsion of the Soviet kind is neither available to a modern state nor desirable, and the evidence says nothing about whether payments for attendance, which many countries use for the schooling of girls, would reach adult women in the same way.

At the same time, there are several things that the evidence does not show. It does not identify the motive of the state, since conditions (2) and (3) produce the same outcome whether the state valued reading citizens or reading mothers. It does not measure what compulsion cost those subjected to it. It also measures literacy as the censuses defined it, that is, the ability to read, which a ten-month course could give and which is far from schooling. Collectivisation changed the village in the same years, and a rise in the private return to a woman's literacy would have worked in the same direction as the campaign. It may be argued, however, that collectivisation is less an alternative to the argument of Engels than a part of it. His remedy was the return of women to public production, and the collective farm, which credited labour to each of its members, was the Soviet form of that return. The campaign and the collective farm removed decisions about the time and the skills of women from the household at the same moment, and the cohort changes observed here include both, together with the other changes of those years.

7 Conclusion

This paper documents a large increase in the recorded literacy of adult female cohorts during the Soviet campaign against illiteracy. The rural cohort aged 20 to 34 in 1926 rose from 43 to 61 per cent literate by 1939, and the later census records sustained high literacy among surviving members. The younger-minus-older female contrast is 20 points under the main specification and remains positive in the reported sample restrictions. Its causal attribution to the campaign depends on restrictions on counterfactual age-specific learning, selection and reporting.

These findings contribute to three fields. For the economics of adult education they document a large cohort literacy change during a major adult education campaign, together with competing interpretations of its causes. For the study of nation-building they extend the reach of the state school to adults, and for the history of the Soviet Union they distinguish recorded within-cohort changes from cohort replacement, subject to selection and comparability of measurement. They also give empirical content to a thesis that the founders of the Soviet state took from Engels. The expansion of women's cohort literacy coincided with a period of intensified public instruction and institutional change. Separating household decision-making, state reach and private returns requires additional evidence.

The evidence has limits that follow from its sources. The censuses report aggregates by region, cohort and sex, so the estimates describe cohorts, and the behaviour of individual persons remains unobserved. The main sample has 18 units. Literacy here is the ability to read as the respondent declared it and the enumerator recorded it, and it was never tested. The cost of compulsion to those subjected to it remains outside its scope.

Several directions for future research follow from these limits, and the sources for them exist. The remaining regional volumes of the census of 1926 would multiply the number of units, the census of 1920 would separate the first Soviet years from the last imperial decades, and additional regional tables of 1959 would extend the follow-up beyond the nine republics examined in Appendix H. I expect that the occupational tables of 1926 and 1939, which would show what the women who learned to read went on to do and whether they entered the industry of the Five-Year Plans, will be the most informative of these sources, since this is the question that Engels would have asked next.

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Online Appendix

The appendix refers to the period between the censuses of 1897 and 1926 as the household regime and to the period between the censuses of 1926 and 1939 as the state regime. The cohorts of the first period are labelled by their age at the decree of December 1919, which is their age at the census of 1897 plus twenty-three years. Section A describes the sources and the validation of the 1926 tables. Section B sets out the model and Section C the estimating equations. Sections D and E report the two periods cohort by cohort, with their robustness checks. Section F reports the regional test of Section 5, and Section G collects further tables and figures.

A Data

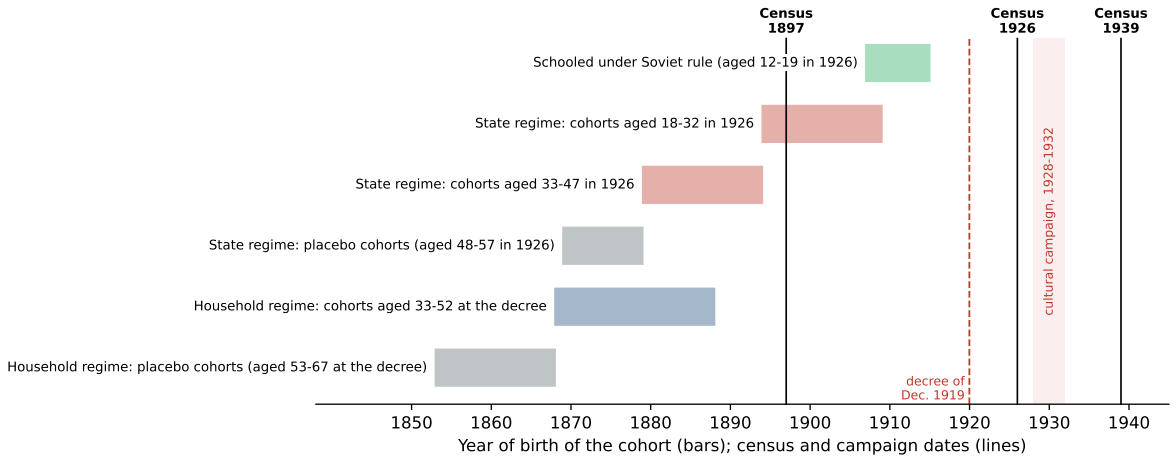
A.1 Census sources and the definition of literacy

The censuses of 1897, 1926 and 1939 define literacy in the same way. The census of 1897 recorded the ability to read. The census of 1926 recorded separately the ability to read, counted if the respondent could read at least by syllables, and the ability to write; in the published tables a literate person is one who can read, for comparability with 1897. The census of 1939 asked whether the respondent reads and writes, or only reads, in any language, and counted both as literate (Tsentral'noe Statisticheskoe Upravlenie SSSR, 1928; Zhiromskaya et al., 1996). The constancy of the definition removes the most obvious mechanical explanation of the gains between 1926 and 1939, and it implies that Arabic-script literacy among Muslim women was counted in 1897 and 1926.[3]

The 1926 census was taken on 17 December 1926 and the 1939 census on 17 January 1939, which is approximated by shifting the age window by twelve years; some respondents are thirteen years older. The census of 1897 was taken on 9 February, twenty-nine years and ten months before that of 1926, so that five-year age bands of 1897 correspond to five-year bands of 1926 shifted by thirty years. Volume 9 of the 1926 census recomputed the 1897 results to the 1927 borders of the nineteen Gosplan regions of the RSFSR, using the territorial coefficients of V. P. Efremov corrected by pre-war *volost* populations; this official recomputation is the basis of the link from 1897 to 1926. Figure A1 shows which birth cohorts are observed in which census and how old they were at the decree of 1919 and at the start of the cultural campaign in 1928.

[3]The 1939 census is known to have been inflated in its population totals (Tolts, 2021; Zhiromskaya, 2001; Andreev et al., 1998). The inflation was of the order of one to two per cent and concerned the count rather than the literacy question. Section E.3 shows that the literacy gains do not have the form a uniform inflation would produce.

Figure A1: Birth cohorts observed across the three censuses



Note: Bars show the birth years of the cohorts used in the analysis; vertical lines mark the census dates; the dashed line marks the decree of December 1919 and the shaded band the cultural campaign of 1928 to 1932.

A.2 The 1926 tables and their validation

The regional volumes of the 1926 census print, for every *guberniya*, *okrug*, autonomous republic and autonomous oblast, a Table IX giving the population by single year of age, sex and literacy, for the whole population and for each of the most numerous nationalities, separately for the whole territory and for its urban and rural parts. These tables had not been digitised. From the scanned volumes held by the Russian State Library I transcribed Table IX for 44 territorial units, 90,642 cells in all. Rural pages are available for 35 of them, and the analyses of the rural population use these. The units are eight autonomous units (the Tatar, Bashkir, Chuvash, Karelian and Volga German republics and the Mari, Votyak and Kalmyk autonomous oblasts), twenty Russian *gubernii*, from Murmansk and Leningrad in the north-west through the Black Earth centre to Astrakhan and Orenburg in the south-east, and the sixteen *okrugi* of the Ural oblast.

The tables carry the means of their own verification. Beside the single years of age they print a subtotal for every five years, so that each five-year figure exists twice, as the printed subtotal and as the sum of five single years transcribed separately. A five-year group enters the analysis if the two agree to within two persons or half of one per cent. Where they disagree, the figure is taken that satisfies one of two further identities of the table, that the urban and rural populations sum to the total and that the nationalities sum to the whole population. Where neither identity discriminates, the figure is taken that lies close to the two adjacent cohorts of the same series while the other does not. Groups that fail all three tests are dropped. Of the 8,840 five-year groups between the ages of 10 and 69, 91.8 per

cent pass the first test, 0.9 per cent the second and 1.0 per cent the third, and 6.3 per cent are dropped. The dropped groups include 39 that pass the identities and differ from both adjacent cohorts by more than 25 points of literacy or by a factor of three in population, a pattern produced by an error that is present in both readings and that an identity cannot detect. Section A.5 describes the procedure in full. The validated data contain 5,984 cells by unit, area, nationality, sex and age band, of which 2,433 are rural. Six units have rural pages on which fewer than 85 per cent of the single-year cells satisfy an identity (Astrakhan, Kungur, the Volga German republic, Orenburg, Penza and Samara); they are excluded in robustness checks.

A.3 The linked cohorts

Table XV of Volume 9 gives, for each of the nineteen Gosplan regions, the population by sex, five-year age band and literacy in 1926 and in 1897. Its 10,320 cells provide the linked cohorts of the household regime and, at the regional level, a replication of the state regime. Of the 532 cells used, 531 satisfy the identity between the urban, rural and total figures. The 1939 tables of literacy by sex, age and urban or rural residence for every region of the RSFSR were declassified from the Russian State Archive of the Economy (fond 1562, opis' 336, dela 607 and 640) and digitised by Demoscope Weekly; I collected them for 75 regions (12,144 cells). They are not by nationality. The table of the 1959 census that gives the population of the RSFSR by age, sex, level of education and literacy, separately for town and village (the same fond and opis', dela 1591 to 1594, also digitised by Demoscope Weekly), is used once, to check the 1939 figures on the same cohorts twenty years later.

A unit of 1926 can be linked to a successor of 1939 in 29 cases, of which 20 have rural pages. The cohorts of 1926 are built from validated single years of age, so that a cohort aged 18 to 22 in 1926 is compared with the group aged 30 to 34 in 1939. Four units whose successors of 1939 cannot be compared with them are excluded from every sample.[4] After validation, a cohort being used only where both of its sexes are validated, the rural sample of the state regime contains 218 cells by unit, cohort and sex from 18 units. The oblasts of 1939 do not have the borders of the *gubernii* of 1926, and the Ural *okrugi*, three of which are in the rural sample, are matched to the larger oblast that contained them. The comparison is between literacy rates, which a change of border affects only if the territory gained or lost differed in literacy, and Section E.3 repeats the analysis on the units whose cohorts changed

[4]The Leningrad and Novgorod *gubernii* were both absorbed into the Leningrad oblast of 1939, together with the Pskov *guberniya*, so that the cohorts of 1939 are two to six times the size of those of 1926. The Nizhny Novgorod *guberniya*, whose successor, the Gorky oblast, does not have its borders, and the Volga German republic have no rural pages in any case.

in size by no more than mortality would imply, without the Ural *okrugi*, and on the seventeen Gosplan regions that cover the whole RSFSR.

A.4 Summary of the data

Table A1 lists the sources and the files built from them, Table A2 describes the three linked samples, and Table A3 reports the transcription and the validation of every unit of 1926. Figure A2 maps the units on the district boundaries of 1897.

Table A1: The data: sources, validated tables and linked samples

Source or file	Content	Units	Cells
A. Sources			
Census of 1926, Table IX of the regional volumes	Population by single year of age, sex, nationality and literacy, total, urban and rural	44	90,642
Census of 1926, Table XV of Volume 9	Population by sex, age and literacy in 1926 and in 1897, borders of 1927	19	10,320
Census of 1939 (RGAE, via Demoscope Weekly)	Population and illiterate by sex, age, urban and rural, by region	75	12,144
Census of 1959 (RGAE, via Demoscope Weekly)	Population by sex, age, education and literacy, RSFSR	1	225
B. Validated tables of 1926			
Five-year groups, ages 10 to 69	Unit, area, nationality, sex		8,840
Single years of age, 8 to 57	Unit, area, nationality, sex		37,596
Age bands	Unit, area, nationality, sex		5,984
C. Linked cohorts			
1926 to 1939, units (main sample)	Rural, cohort by sex	18	218
1926 to 1939 and 1959, regions	Rural, cohort by sex	17	238
1897 to 1926, regions	Rural, cohort by sex	19	265

Note: Cells are the figures of the source or the rows of the file built from it. RGAE is the Russian State Archive of the Economy (fond 1562, opis' 336).

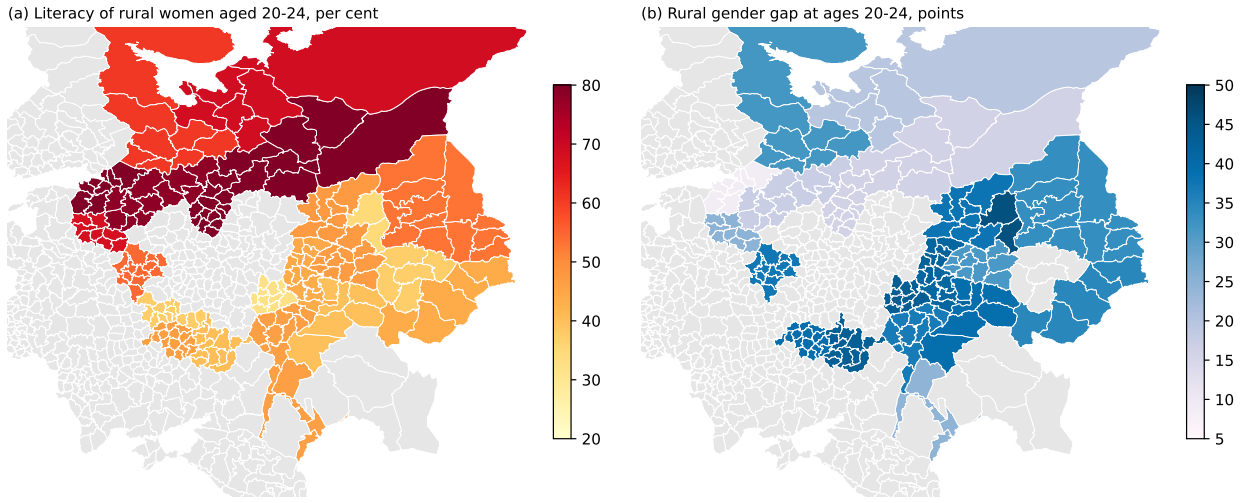
Figure A3 shows the age and sex profile of rural literacy in 1926 pooled over the units. Men's literacy peaks at 84 per cent in the cohorts aged 20 to 29, who were of school age before 1917 and of conscription age during the wars; women's peaks at 47 per cent at ages 20 to 24 and falls to 14 per cent at ages 40 to 49. The gender gap is 16 points among children aged 8 to 11, 37 points at 20 to 24 and 50 points at 30 to 39.

Table A2: Summary statistics of the linked samples, rural population

			Literate, per cent		Change, points			Survival
Sample	Sex	Cells	earlier	later	mean	SD	range	ratio
1926 to 1939, units	women	109	24.0	39.0	15.0	9.7	1 to 37	0.85
	men	109	67.7	76.5	8.9	6.1	−5 to 29	0.76
1926 to 1939, regions	women	119	24.2	37.3	13.1	9.8	1 to 55	0.75
	men	119	63.9	74.2	10.3	10.7	0 to 54	0.66
1897 to 1926, regions	women	132	8.9	7.9	−1.0	2.6	−8 to 7	0.53
	men	133	32.0	38.2	6.2	9.9	−10 to 36	0.47

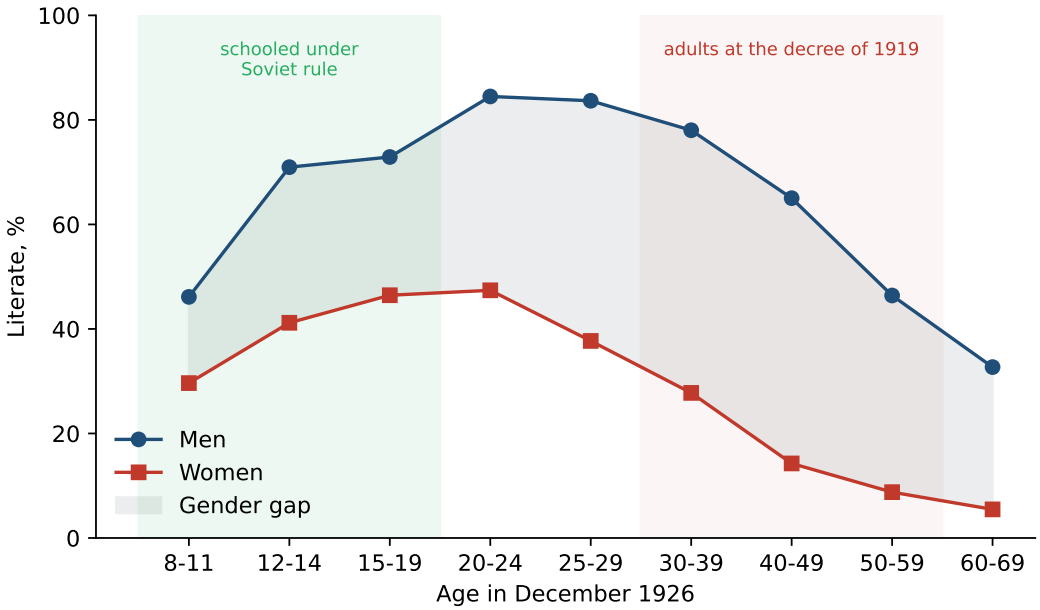
Note: Unweighted means over the cells of each sample, all cohorts including those above the age limit. A cell is a cohort of one sex in one unit or region. The survival ratio is the median of the number of the cohort at the later census divided by its number at the earlier census.

Figure A2: The units of the 1926 tables: women's literacy and the gender gap at ages 20 to 24, rural, 1926



Note: District boundaries of 1897 from Malein (2025); districts are coloured by the value of the 1926 unit to which they belong (provinces divided between several units carry the mean of those units). Grey districts are outside the sample or have no validated cell. Panel (a): literacy of rural women aged 20 to 24, all nationalities; panel (b): male minus female literacy at the same ages.

Figure A3: Literacy by age and sex in the 1926 census, rural population



Note: Population-weighted over the units with validated rural cells, all nationalities. The green band marks cohorts schooled entirely under Soviet rule and the red band cohorts that were adults at the decree of 1919.

A.5 Transcription and validation of the 1926 tables

The tables were transcribed by hand from the scans of the regional volumes held by the Russian State Library, page by page, into a spreadsheet with one row per single year of age and one column per nationality, sex and literacy status. The transcription was then checked in two ways. First, an automatic reading of every page was obtained with a neural recogniser, and each hand-entered cell was compared with the machine reading of the same cell. Cells on which the two disagreed were re-examined against the scan. Second, every figure was tested against the arithmetic of the tables themselves, in the following order.

Figure A4 shows a page of the source. Each page gives, for one unit and one area, the population and the literate by single year of age and sex for the whole population and for each nationality, and every fifth row, printed in bold, is the subtotal of the five-year group above it.

The printed subtotal. Table IX prints, after every five single years of age, the subtotal of the five-year group. For each unit, area (total, urban or rural), nationality and sex, and separately for the population and for the literate, the printed subtotal P is compared with the sum S of the five single years. If they agree to within two persons or half of one per cent, the group is validated by the first test.

The identities of the table. If P and S disagree, or one of them is illegible, two further

Figure A4: A page of Table IX of the census of 1926: the Tatar republic, rural areas

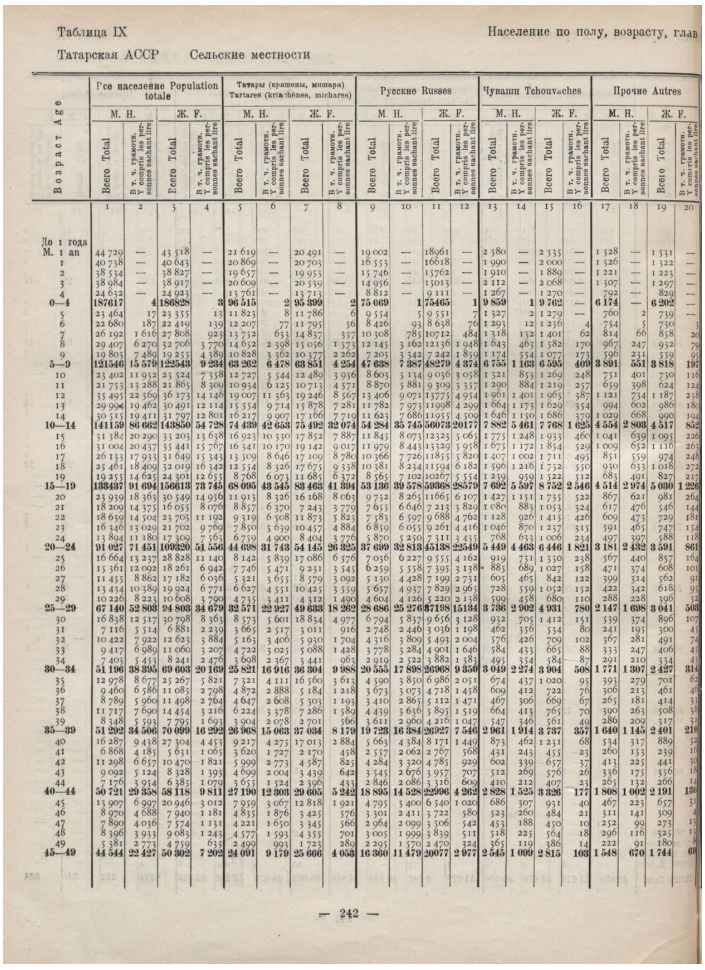


Таблица IX

Татарская АССР Сельские местности

Население по полу, возрасту, гла

Возраст Age	Все население Population totale				Татары (крящены, мишари) Tartares (kriachènes, michares)				Русские Russes				Чуваши Tchouvaches				Прочие Autres			
	М. Н.		Ж. F.		М. Н.		Ж. F.		М. Н.		Ж. F.		М. Н.		Ж. F.		М. Н.		Ж. F.	
	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
До 1 года																				
М. 1 an	44 729	—	43 518	—	21 619	—	20 491	—	19 002	—	18061	—	2 380	—	2 535	—	1 528	—	1 531	—
1	40 738	—	40 643	—	20 869	—	20 703	—	16 553	—	16618	—	1 990	—	2 000	—	1 326	—	1 122	—
2	38 534	—	38 827	—	19 657	—	19 953	—	15 746	—	15762	—	1 910	—	1 889	—	1 221	—	1 223	—
3	38 984	—	38 917	—	20 009	—	20 539	—	14 956	—	15013	—	2 112	—	2 068	—	1 307	—	1 297	—
4	24 612	—	24 923	—	13 761	—	13 713	—	8 812	—	9 111	—	1 267	—	1 270	—	792	—	829	—
0—4	**187617**	**4**	**186828**	**3**	**96 515**	**2**	**95 399**	**2**	**75 069**	**1**	**75465**	**1**	**9 859**	**1**	**9 762**	**—**	**6 174**	**—**	**6 202**	**—**
5	23 464	17	23 355	13	11 823	8	11 786	6	9 554	5	9 551	7	1 327	2	1 279	—	760	2	739	—
6	22 680	187	22 419	139	12 207	77	11 795	56	8 426	93	8638	76	1 293	12	1 258	4	754	5	730	1
7	26 192	1 616	27 808	923	13 752	633	14 837	357	10 308	785	10712	484	1 318	152	1 401	62	814	66	858	20
8	29 407	6 270	32 706	3 770	14 652	2 398	15 016	1 573	12 145	3 162	12136	1 948	1 643	463	1 582	170	967	247	932	79
9	19 803	7 489	19 255	4 389	10 828	3 362	10 377	2 262	7 205	3 342	7 242	1 859	1 174	554	1 077	173	596	231	559	95
5—9	**121546**	**15 579**	**122543**	**9 234**	**63 262**	**6 478**	**63 851**	**4 254**	**47 638**	**7 387**	**48279**	**4 374**	**6 755**	**1 183**	**6 595**	**409**	**3 891**	**561**	**3 818**	**197**
10	23 402	11 932	23 524	7 358	12 727	5 544	12 489	3 936	8 603	5 134	9 036	3 058	1 321	853	1 269	248	751	401	730	116
11	21 753	13 288	21 865	8 309	10 934	6 125	10 713	4 571	8 870	5 881	9 309	3 357	1 290	884	1 219	257	659	398	624	124
12	35 495	22 569	36 173	14 146	19 007	11 363	19 246	8 567	13 406	9 071	13775	4 954	1 961	1 401	1 965	587	1 121	734	1 187	218
13	29 998	19 462	30 491	12 114	15 554	9 714	15 878	7 281	11 782	7 973	11998	4 299	1 664	1 173	1 629	354	994	602	986	188
14	30 515	19 411	31 797	12 801	16 217	9 907	17 166	7 719	11 623	7 686	11955	4 509	1 646	1 150	1 686	379	1 029	668	990	194
10—14	**141159**	**86 662**	**143850**	**54 728**	**74 439**	**42 653**	**75 492**	**32 074**	**54 284**	**35 745**	**56073**	**20177**	**7 882**	**5 461**	**7 768**	**1 625**	**4 554**	**2 803**	**4 517**	**852**
15	31 582	20 296	33 201	13 618	16 021	10 330	17 812	7 887	11 845	8 073	12121	5 063	1 773	1 248	1 933	456	1 041	639	1 095	226
16	31 004	20 457	33 441	15 767	16 341	10 570	19 142	9 017	11 979	8 443	13329	5 938	1 673	1 172	1 854	529	1 009	672	1 116	263
17	26 133	17 933	31 649	15 343	13 509	8 646	17 109	8 780	10 366	7 726	11855	5 820	1 407	1 002	1 711	495	851	559	974	248
18	25 401	18 409	32 019	16 342	12 554	8 326	17 675	9 338	10 381	8 254	13594	6 182	1 596	1 216	1 732	550	930	653	1 018	272
19	19 253	14 693	24 301	12 655	8 768	6 073	11 685	6 372	8 563	7 102	10267	5 514	1 239	959	1 322	512	683	491	827	217
15—19	**133437**	**91 694**	**156618**	**78 745**	**68 095**	**43 545**	**83 463**	**41 394**	**53 136**	**39 578**	**59308**	**28579**	**7 692**	**5 597**	**8 752**	**2 546**	**4 514**	**2 974**	**5 030**	**1 226**
20	23 939	18 363	30 549	14 956	11 913	8 326	16 168	8 063	9 732	8 265	11665	6 107	1 427	1 151	1 735	522	867	621	981	264
21	18 209	14 375	16 051	8 076	8 857	6 370	7 243	3 779	7 651	6 646	7 213	3 829	1 080	881	1 053	324	617	476	546	144
22	18 639	14 504	23 793	11 192	9 319	6 508	11 873	5 823	7 383	6 597	9 688	4 762	1 128	926	1 415	426	669	473	729	181
23	16 346	13 029	21 702	9 769	7 850	5 639	10 457	4 884	6 859	6 055	9 261	4 416	1 046	870	1 237	315	591	465	747	154
24	11 894	11 180	17 309	7 563	6 755	4 900	8 404	3 775	4 870	4 350	7 311	3 433	788	633	1 006	234	497	397	588	118
20—24	**91 027**	**71 451**	**109320**	**51 556**	**44 698**	**31 743**	**54 145**	**26 325**	**37 499**	**32 813**	**45138**	**22549**	**5 449**	**4 463**	**6 446**	**1 821**	**3 181**	**2 432**	**3 591**	**801**
25	16 664	13 237	28 828	11 140	8 142	5 839	17 086	6 376	7 036	6 227	9 555	4 162	919	731	1 330	238	567	440	857	164
26	15 361	12 092	18 261	6 942	7 746	5 471	9 211	3 545	6 259	5 518	7 395	3 138	885	689	1 027	158	471	374	608	101
27	11 455	8 862	17 182	6 036	5 321	3 653	8 579	3 092	5 130	4 428	7 199	2 731	605	465	842	122	399	314	562	91
28	13 434	10 389	19 924	6 771	6 627	4 551	10 435	3 559	5 657	4 937	7 829	2 965	728	559	1 032	132	422	342	618	95
29	10 226	8 223	10 608	3 790	4 735	3 411	4 312	1 490	4 604	4 126	5 220	2 138	599	458	680	110	288	228	396	52
25—29	**67 140**	**52 803**	**94 803**	**34 679**	**32 571**	**22 927**	**49 633**	**18 202**	**28 686**	**25 276**	**37198**	**15194**	**3 736**	**2 902**	**4 931**	**780**	**2 147**	**1 698**	**3 041**	**503**
30	16 858	12 517	30 708	8 163	8 573	5 601	18 814	4 977	6 794	5 837	9 616	3 128	932	705	1 412	151	539	374	896	107
31	7 116	5 514	6 881	2 219	3 663	2 517	3 014	916	2 748	2 446	3 036	1 198	462	356	534	80	241	195	300	41
32	10 422	7 922	12 623	3 884	5 163	3 406	5 930	1 704	4 316	3 809	5 493	2 004	576	426	709	102	367	281	491	74
33	9 417	6 989	11 066	3 207	4 722	3 025	5 088	1 428	3 778	3 284	4 901	1 646	584	433	665	88	333	247	406	45
34	7 403	5 453	8 241	2 456	3 698	2 367	3 441	961	3 019	2 522	3 882	1 183	493	358	584	87	293	210	334	43
30—34	**51 196**	**38 395**	**69 603**	**20 169**	**25 821**	**16 916**	**36 304**	**9 988**	**20 555**	**17 898**	**26908**	**9 350**	**3 049**	**2 274**	**3 904**	**508**	**1 771**	**1 307**	**2 427**	**310**
35	12 978	8 677	25 267	5 821	7 321	4 111	16 560	3 613	4 590	3 850	6 986	2 051	674	437	1 020	93	393	279	701	62
36	9 460	6 186	11 083	2 798	4 872	2 888	5 184	1 218	3 673	3 073	4 718	1 458	609	412	722	76	306	213	461	41
37	8 789	5 969	11 498	2 764	4 647	2 608	5 303	1 193	3 410	2 865	5 112	1 471	467	306	669	67	265	181	414	33
38	11 717	7 690	14 454	3 216	6 224	3 378	7 286	1 589	4 439	3 636	5 895	1 519	664	413	765	70	390	263	508	38
39	8 348	5 593	7 795	1 693	3 904	2 078	2 701	866	3 611	2 960	4 216	1 047	547	346	561	49	286	209	337	30
35—39	**51 292**	**34 506**	**70 099**	**16 292**	**26 968**	**15 063**	**37 034**	**8 179**	**19 723**	**16 384**	**26927**	**7 546**	**2 961**	**1 914**	**3 737**	**357**	**1 640**	**1 145**	**2 401**	**210**
40	16 287	9 418	27 304	4 435	9 237	4 973	17 051	2 884	5 662	4 584	8 171	1 449	873	462	1 233	68	534	317	869	52
41	6 868	4 185	5 631	1 063	3 620	1 727	2 170	458	2 557	2 062	2 767	568	431	243	455	23	260	153	239	10
42	11 298	6 657	10 470	1 821	5 999	2 773	4 587	823	4 284	3 320	4 785	929	602	339	657	37	413	225	441	32
43	9 092	5 324	8 328	1 393	4 699	2 004	3 459	642	3 543	2 676	3 917	707	511	269	576	26	339	175	376	8
44	7 776	3 914	6 381	1 079	3 653	1 324	2 376	431	2 846	2 086	3 316	669	410	218	407	23	265	132	266	14
40—44	**50 721**	**29 358**	**58 118**	**9 811**	**27 190**	**12 803**	**29 605**	**5 242**	**18 895**	**14 528**	**22996**	**4 262**	**2 828**	**1 525**	**3 328**	**177**	**1 808**	**1 002**	**2 191**	**136**
45	13 907	6 997	20 946	3 012	7 939	3 667	12 818	1 921	4 795	3 400	6 540	1 020	686	307	951	49	467	223	657	31
46	8 970	4 888	7 940	1 181	4 833	1 876	3 451	576	3 301	2 411	3 722	586	583	260	484	21	311	141	309	8
47	7 890	4 036	7 574	1 131	4 221	1 650	3 345	566	2 964	2 099	3 506	542	453	188	450	10	252	99	273	13
48	8 396	3 933	9 083	1 243	4 577	1 593	4 355	701	3 005	1 999	3 859	511	518	225	564	18	296	116	325	15
49	4 381	2 731	4 759	633	2 499	993	2 723	288	2 291	1 579	2 470	324	365	119	386	14	222	91	186	9
45—49	**44 544**	**22 427**	**50 302**	**7 202**	**24 091**	**9 179**	**25 666**	**4 053**	**16 366**	**11 479**	**20077**	**2 977**	**2 545**	**1 099**	**2 815**	**103**	**1 548**	**670**	**1 744**	**69**

— 242 —

(a) The page

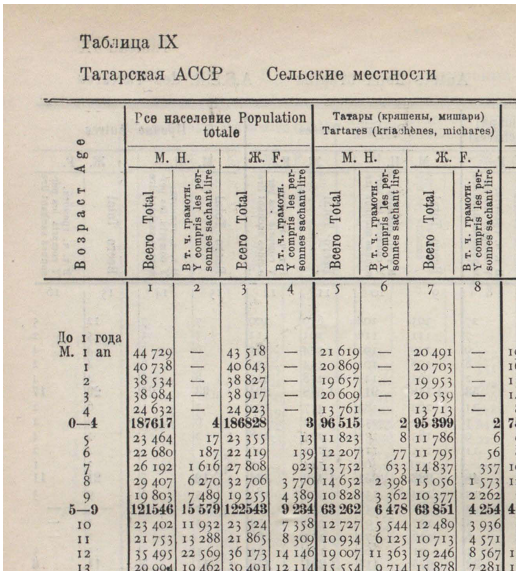


Таблица IX

Татарская АССР Сельские местности

Возраст Age	Все население Population totale				Татары (крящены, мишари) Tartares (kriachènes, michares)			
	М. Н.		Ж. F.		М. Н.		Ж. F.	
	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire	Всего Total	В т. ч. грамотн. Y compris les personnes sachant lire
	1	2	3	4	5	6	7	8
До 1 года								
М. 1 an	44 729	—	43 518	—	21 619	—	20 491	—
1	40 738	—	40 643	—	20 869	—	20 703	—
2	38 534	—	38 827	—	19 657	—	19 953	—
3	38 984	—	38 917	—	20 009	—	20 539	—
4	24 632	—	24 923	—	13 761	—	13 713	—
0—4	**187617**	**4**	**186828**	**3**	**96 515**	**2**	**95 399**	**2**
5	23 464	17	23 355	13	11 823	8	11 786	6
6	22 680	187	22 419	139	12 207	77	11 795	56
7	26 192	1 616	27 808	923	13 752	633	14 837	357
8	29 407	6 270	32 706	3 770	14 652	2 398	15 036	1 573
9	19 803	7 489	19 255	4 389	10 828	3 362	10 377	2 262
5—9	**1215546**	**15 579**	**122543**	**9 234**	**63 262**	**6 478**	**63 851**	**4 254**
10	23 402	11 932	23 524	7 358	12 727	5 544	12 489	3 936
11	21 753	13 288	21 865	8 309	10 934	6 125	10 713	4 571
12	35 495	22 569	36 173	14 146	19 007	11 363	19 246	8 567
13	29 998	19 462	30 491	12 114	15 554	9 714	15 878	7 281
14								

(b) Detail, ages 0 to 14

Note: Volume 3 of the census of 1926, page 242, from the scan of the Russian State Library. The rows 0–4, 5–9 and 10–14 in bold are the printed subtotals against which the sums of the single years are checked.

identities are used, computed from groups that have passed the first test. The urban and rural figures must sum to the total, and the nationalities must sum to the whole population. If exactly one of P and S satisfies an identity, it is taken. If both are missing and the two identities agree with each other, their value is taken.

The adjacent cohorts. If no identity discriminates, P and S are compared with the mean of the two adjacent five-year groups of the same series. The one is taken that lies within 15 points of literacy of that mean, or within 40 per cent of it for the population, provided that the other does not.

A group that fails the three tests is dropped. A single year of age is used only if its five-year group has been validated and the validated figure is the sum of the single years. An error that was made in the printed subtotal and in a single year alike, as happens when a

digit of the printed page is misread consistently, passes every identity. A last rule therefore drops the groups of at least 100 persons whose literacy differs by more than 25 points from both adjacent groups, and those whose population is more than three times, or less than a third of, both adjacent groups. The age bands used in the analysis are sums of validated groups, or of validated single years for the bands 8 to 11 and 12 to 14 and for the cohorts linked to 1939. Bands with fewer than 200 persons, series with fewer than three bands, and bands that differ by more than 30 points from both adjacent bands are dropped.

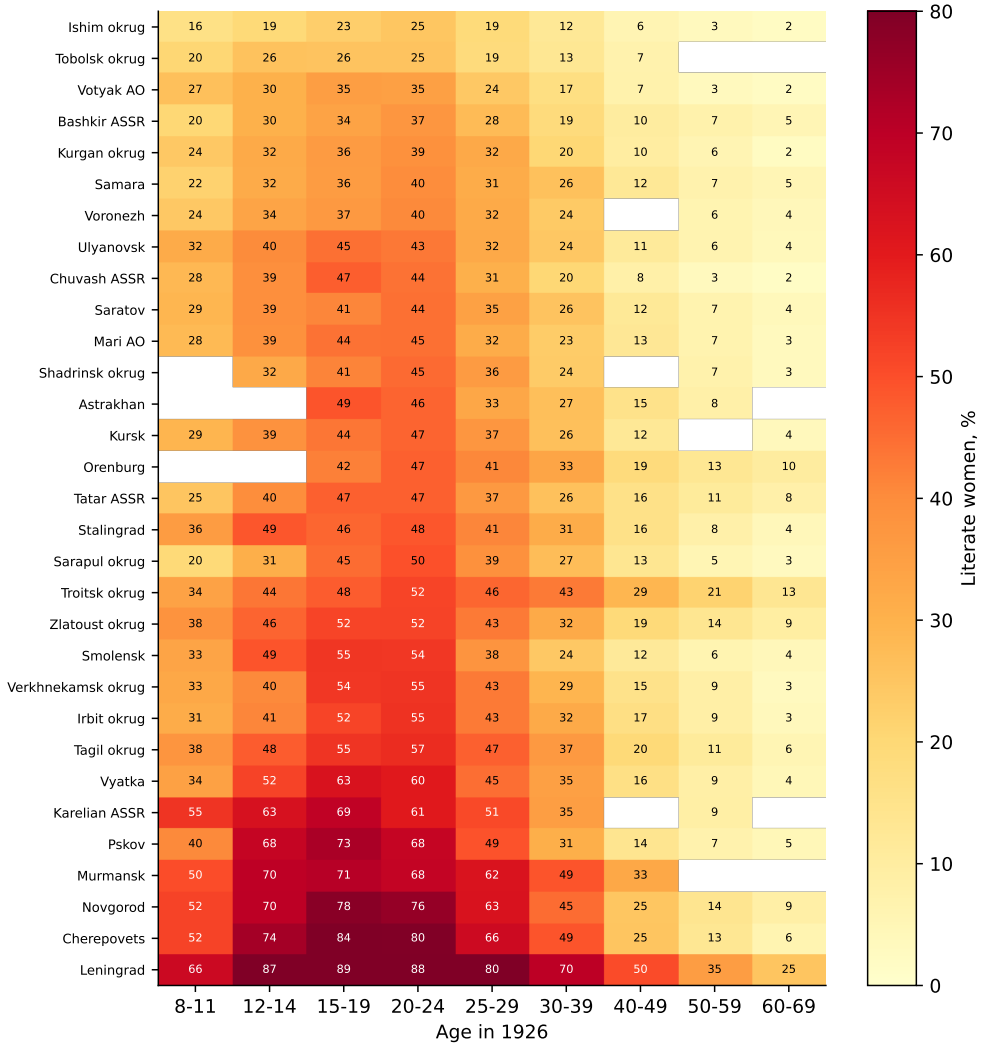
Of the 8,840 five-year groups between the ages of 10 and 69, 91.8 per cent are validated by the first test, 0.9 per cent by the second and 1.0 per cent by the third. In the rural population the shares are 90.1, 0.7 and 1.2 per cent of 3,624 groups. Of the 37,596 single-year cells between the ages of 8 and 57, 92.9 per cent are validated. The results of the paper are unchanged when only the figures validated by the first two tests are used (Table A10). Figure A5 shows the literacy of rural women by unit and cohort in the validated data.

Table A3: The units of Table IX of the census of 1926: transcription and validation

Unit	Cells transcribed	Nation-alities	Rural pages	Five-year groups	Validated, per cent		Main sample
					first test	all tests	
Autonomous republics							
Bashkir ASSR	4,428	8	yes	432	98.4	98.8	•
Chuvash ASSR	1,968	3	yes	192	99.5	100.0	•
Karelian ASSR	1,968	3	yes	192	95.8	95.8	•
Tatar ASSR	2,460	4	yes	240	100.0	100.0	•
Volga German ASSR	1,830	4	yes	200	73.0	78.0	
Autonomous oblasts							
Kalmyk AO	360	2	–	48	97.9	100.0	
Mari AO	1,968	3	yes	192	94.8	94.8	•
Votyak AO	1,968	3	yes	192	97.4	99.5	•
Gubernii							
Astrakhan	1,830	4	yes	200	75.0	78.0	
Bryansk	2,460	4	yes	240	78.3	80.8	
Cherepovets	2,214	2	yes	216	100.0	100.0	
Kursk	1,968	3	yes	192	96.9	97.4	•
Leningrad	2,952	3	yes	96	96.9	99.0	
Murmansk	1,080	2	yes	144	100.0	100.0	
Nizhny Novgorod	1,230	4	–	120	76.7	76.7	
Novgorod	2,214	2	yes	216	97.7	99.5	
Orel	738	2	–	72	100.0	100.0	
Orenburg	2,460	4	yes	240	85.0	86.2	•
Penza	2,460	4	yes	240	83.8	87.5	•
Pskov	2,214	2	yes	216	98.1	100.0	
Samara	3,444	6	yes	336	81.8	83.6	•
Saratov	3,444	6	yes	336	98.2	99.1	•
Smolensk	1,968	3	yes	192	89.1	93.2	•
Stalingrad	1,968	3	yes	192	97.9	98.4	•
Tambov	738	2	–	72	94.4	97.2	
Ulyanovsk	2,196	5	yes	240	97.9	99.2	
Voronezh	1,968	3	yes	192	90.1	93.8	•
Vyatka	1,968	3	yes	192	97.9	99.0	•
Okrugi *of the Ural oblast*							
Chelyabinsk okrug	984	3	–	96	96.9	97.9	
Irbit okrug	2,952	3	yes	288	85.4	92.0	
Ishim okrug	2,952	3	yes	288	92.0	97.2	•
Komi-Permyak okrug	984	3	–	96	82.3	83.3	
Kungur okrug	1,968	3	yes	192	93.2	93.2	
Kurgan okrug	2,952	3	yes	288	95.1	96.5	•
Perm okrug	984	3	–	96	89.6	92.7	
Sarapul okrug	1,968	3	yes	192	83.3	83.3	
Shadrinsk okrug	2,952	3	yes	288	87.8	91.3	
Sverdlovsk okrug	480	3	–	64	100.0	100.0	
Tagil okrug	1,968	3	yes	192	97.9	97.9	
Tobolsk okrug	1,440	3	yes	192	92.2	97.9	•
Troitsk okrug	1,968	3	yes	192	97.9	99.0	
Tyumen okrug	984	3	–	96	96.9	97.9	
Verkhnekamsk okrug	2,952	3	yes	288	93.8	95.8	
Zlatoust okrug	2,952	3	yes	288	79.5	82.3	
All 44 units	89,904		35	8,768	91.7	93.6	18

Note: Cells transcribed are all figures of the unit, for the whole territory and for its urban and rural parts. Nationalities are the groups printed in the table, including the group of other nationalities. Five-year groups are those between the ages of 10 and 69 by area, nationality and sex, and the validated shares are those passing the first test and any of the three tests. The main sample marks the units linked to 1939 with validated rural pages. The table of the Ural oblast as a whole, the sum of its *okrugi*, was transcribed as a check and is not shown.

Figure A5: Women's literacy by unit and cohort, rural, 1926



Note: All nationalities, validated cells; blank cells did not pass validation. Units with validated cells in the four bands between the ages of 15 and 39.

B A Model of Household and State Investment in Literacy

The model states the conditions under which a household and a state teach an adult to read, and derives from the difference between them the predictions tested in Section 5.

B.1 The household

A household consists of members indexed by sex $s \in \{m, f\}$ and age a. Teaching a member to read costs c in foregone labour and effort, and teaching a woman costs in addition ϕn, the care of the n young children of the household that she would forgo. A literate member of sex s earns a private return r_s per period from age a until the end of working life T. The

household discounts at rate ρ, so that the present value of the private return is $r_s A(T-a)$, where $A(x) = \sum_{t=1}^{x}(1+\rho)^{-t}$. A literate woman also raises the human capital of her children, worth γ. The household values this only through altruism $\alpha < 1$ and receives it a generation later, after τ periods. The household teaches member (s, a) if

$$V_s(a,n) \;\equiv\; r_s\, A(T-a) \;+\; \mathbf{1}\{s=f\}\left[\frac{\alpha\gamma}{(1+\rho)^{\tau}} - \phi n\right] \;\geq\; c. \tag{A1}$$

This is the condition of Ben-Porath (1967) with a sex-specific return, a discounted intergenerational term and a cost of time. The household is treated as a single decision-maker. If the decision belongs to its head, his valuation of the literacy of a wife or a daughter takes the place of the term in brackets and the condition keeps its form (Doepke and Tertilt, 2009). Instruction that reaches a member outside the household and at no cost to it, as the army's reached the conscript, enters as a lower c for sons. Let the cost c vary across households with distribution function H. The share of members of sex s and age a who are taught is then $H\Big(V_s(a,n)\Big)$.

Three properties follow. First, the share taught declines with age for each sex, since $A(T-a)$ falls in a, and for men it declines from a high level, since r_m is large. Second, when r_f is close to zero, as it was for rural women whose labour was on the family plot, V_f has no private-return age gradient only when fertility, child benefits, opportunity costs and the cost distribution are also held fixed. Low private returns alone do not imply zero learning or an age-invariant female profile. Third, anything that raises the private return r_f, such as the prospect of promotion that literacy opened in the 1930s, raises the share of women taught by the household, and raises it most where the return rises most. This is the incentive explanation of Section 5.

B.2 The state

The state values the same literacy at its social return. This comprises the private return, the intergenerational return γ undiscounted by altruism and by the generation of delay, and a political return π from a reading citizen. Its cost of teaching an adult of age a is $c_S(a)$, which rises with age because older adults learn more slowly, and which varies across adults with distribution function K_a. The state teaches member (s, a) if

$$r_s\, A(T-a) \;+\; \mathbf{1}\{s=f\}\,\gamma \;+\; \pi \;\geq\; c_S(a), \tag{A2}$$

so that the share of the illiterate of sex s and age a whom its instruments reach is $e_s(a) = K_a\Big(r_s A(T-a) + \mathbf{1}\{s=f\}\gamma + \pi\Big)$, which falls with age only under restrictions on the age

distribution of costs and benefits. It is close to a common $e(a) = K_a(\pi)$ if the political return, which does not depend on sex or on local opportunities, is large relative to the returns that do, as the scale of the campaign suggests. Those whom the state does not reach remain under condition (A1), so that under the state regime the share of the illiterate of sex s and age a who are taught is

$$P_s(a) \;=\; e_s(a) + \Big(1 - e_s(a)\Big)\, H\Big(V_s(a,n) \mid \text{not reached by the state}\Big). \tag{A3}$$

Here the household probability is conditional on not being reached. Replacing it by an unconditional probability from an earlier period requires a transportability assumption, equal exposure horizons or a specified hazard model, and restrictions on selection into state instruction. The household component, the second term of equation (A3), is present for men and close to zero for women.

B.3 Predictions

A campaign that works through the household, by offering classes that adults may attend, selects learners by condition (A1). A campaign that works around the household, by enforcing attendance and bearing its cost, selects by equation (A3). The two regimes differ in observable ways.

H1 (*Sex composition.*) Under the household regime the share of the illiterate who are taught is large for men and close to zero for women. Under the state regime it rises for both sexes, from zero to the state's reach for women. The share of men exceeds that of women by the household component of the men whom the state does not reach, and by any excess of the state's reach among men.

H2 (*Age.*) Under the household regime the share of illiterate women who are taught is close to zero at every adult age while that of men declines with age. Under the state regime the shares of both sexes decline with age, since the state's reach does. The excess of men over women, as a share of the men whom a reach equal to that of women would leave out, $(P_m - P_f)/(1 - P_f)$, equals the household component when the reach is common to the sexes, and it declines with age as the share of men taught under the household regime does.

H3 (*Local returns.*) Under the state regime the share of illiterate women who are taught does not vary across regions with the local return to literacy. A rise in that return working through the household, as under the incentive explanation, would raise it where the return rose most.

The model takes the change of regime as given. It is the state's response to the outcome of the first decade, and what the model explains is the difference in outcomes between the two regimes, not the state's decision to change them.

C Estimating Equations and Inference

The Greek letters of the estimating equations in this section denote coefficients and fixed effects and are not the parameters of Section B.

C.1 Linked cohorts and placebo cohorts

The change in a birth cohort's literacy between two censuses is the sum of learning and of selection by mortality and migration. The age limit motivates comparison cohorts with less intended exposure. It does not establish zero exposure or identical age-specific selection. For the household regime I link five-year age bands of 1897 to the same bands thirty years later in 1926, for the rural population of nineteen regions, and estimate

$$\Delta y_{rbs} \;=\; \alpha_r \;+\; \sum_{b \neq b_0} \beta_b D_b \;+\; \gamma F_s \;+\; \sum_{b \neq b_0} \delta_b \, D_b F_s \;+\; \varepsilon_{rbs}, \tag{A4}$$

where Δy_{rbs} is the change in literacy of cohort b of sex s in region r, D_b indicates the cohort's age at the decree in five-year bins from 33-37 to 58-62, F_s indicates women, α_r are region fixed effects, and the reference cohort b_0 was aged 63 to 67 at the decree. The cohorts aged 53 and over at the decree were above its age limit and, as Section D.1 shows, beyond the ages at which any institution of the period taught adults; they serve as comparison cohorts whose change in literacy measures selection alone. The coefficients β_b are difference-in-differences estimates for men of cohort b relative to the reference cohort, and δ_b are triple-difference estimates of the difference between women and men. For the state regime the same specification links the cohorts of 1926 to the corresponding age groups of 1939, with unit fixed effects, cohorts aged 18-22 to 48-52 in 1926 and the cohort aged 53 to 57 as reference. The cohorts aged 48 and over in 1926 had passed the age of 50 when the cultural campaign began and serve as older comparison cohorts; some had been eligible under the 1919 decree, and exact non-exposure during the mass campaign is not observed. A pooled specification replaces the cohort indicators by an indicator of the cohorts that the regime could reach, aged 33 to 47 at the decree in the first period and 18 to 32 in 1926 in the second.

Observations are weighted by the cohort's population in the earlier census and standard errors are clustered by region or by the shared 1939 source territory. The identifying assumption of the triple difference is that the difference between the sexes in selection by mortality

and migration is the same in the cohorts that the regime could reach as in the comparison cohorts. The comparison cohorts allow a check, since under the assumption the difference between the sexes should not vary among them, and Sections D.2 and E.3 bound the bias that selective mortality could produce in the other cohorts.

A change in percentage points depends on the room left below 100 per cent, which differed between the sexes. I therefore estimate the same specifications for a second outcome, the share of the illiterate who were taught,

$$t_{rbs} \;=\; \frac{\Delta y_{rbs}}{100 - y^{0}_{rbs}}, \tag{A5}$$

where y^0 is the literacy of the cohort at the earlier census, with the number of its illiterate as weights. It is the empirical counterpart of H in condition (A1) and of $P_s(a)$ in equation (A3). It equals the share of the illiterate who learned if the cohort is closed and no one who could read ceased to, and selection by mortality and migration enters it as it enters the change in points.

C.2 Inference

Standard errors are clustered by region or unit, with the small-sample correction and the t distribution with one degree of freedom fewer than the number of clusters. Because the clusters are few, 19 regions and 18 units, every main estimate is accompanied by leave-one-out ranges, by a permutation test that reassigns the sexes at random within unit and cohort, and by a wild cluster bootstrap with Rademacher weights (Cameron et al., 2008). Every coefficient and standard error in the tables was estimated independently in two statistical packages, Python and Stata, and the two agree to the last digit.

D The Household Regime, 1897 to 1926

D.1 Adult literacy gains

Table A4 reports the estimates of equation (A4) and Figure A6(a) plots them. Relative to the cohort aged 63 to 67 at the decree, rural men aged 33 to 37 gained 25 points of literacy between 1897 and 1926, men aged 38 to 42 gained 18, men aged 43 to 47 gained 14, and men aged 48 to 52, at the edge of the decree's range, gained 8. Men aged 53 to 57 and 58 to 62 gained 2 to 3 points, which measures the difference in selection between them and the reference cohort. The gradient is monotone at about one point per year of age and shows no break at the age limit of the decree, so that the institutions of the period, the school, the

army and the literacy point, reached men up to the same ages whether or not the decree addressed them. Women of the same cohorts gained 4 points in the youngest cohort and one point or less in every other, relative to the reference cohort. In the pooled specification the men aged 33 to 47 gained 18.3 points (standard error 1.4) relative to the cohorts above the age limit and the women 2.0 (0.4), one ninth as much, and the triple difference is −16.3 points (1.5).[5]

The same contrast appears in the share of the illiterate who were taught (Table A6, panel A, and Figure 2). Of the rural men aged 33 to 47 at the decree who had been illiterate in 1897, 30 per cent could read in 1926. Of the women, who were 62 per cent of the illiterate of these cohorts, the share is 0. The share of men falls from 40 per cent in the youngest cohort to 8 per cent at ages 48 to 52 and to nothing above the age limit. The share of women is between −2 and 3 per cent at every age.

The gains of men are large by the standards of the adult literacy literature. They were made across three decades, and a part of them was made in the army. Conscription took a share of every cohort of men in peacetime and a far larger share between 1914 and 1920 (Castañeda Dower and Markevich, 2018), and both the imperial army and the Red Army, in which an order of September 1919 required classes for the illiterate in every company, taught their recruits to read (Brooks, 1985; Kenez, 1985; Bol'shaya sovetskaya entsiklopediya, 1938). The gains of men decline from cohort to cohort in step with exposure to military service after 1897. The cohorts above the age limit of the decree, which show none, had completed their active service before that year and were too old to be mobilised in 1914.[6] The data cannot divide the gains of men between the army, the school and the workplace, and the argument does not require the division. None of these routes was open to a woman, for whom the only way to literacy was a decision taken at home. In the towns, where more men could read to begin with, the gains of men are smaller and the difference between women and men has the same sign (Table A9).

D.2 Robustness

The estimates of Table A4 are unchanged when unweighted, when the survival ratio is added as a control (its coefficient is −4.0 with a standard error of 7.2), in the total population, and when the five Muslim regions, where Arabic-script literacy could confound the 1897 baseline, are excluded (Table A9). In the cohorts above the age limit the difference between

[5]The youngest cohort was aged 10 to 14 in 1897 and may have completed its schooling after that census. Without it, men aged 38 to 47 gained 14.5 points (1.1) relative to the cohorts above the age limit and the triple difference is −13.9 (1.2).

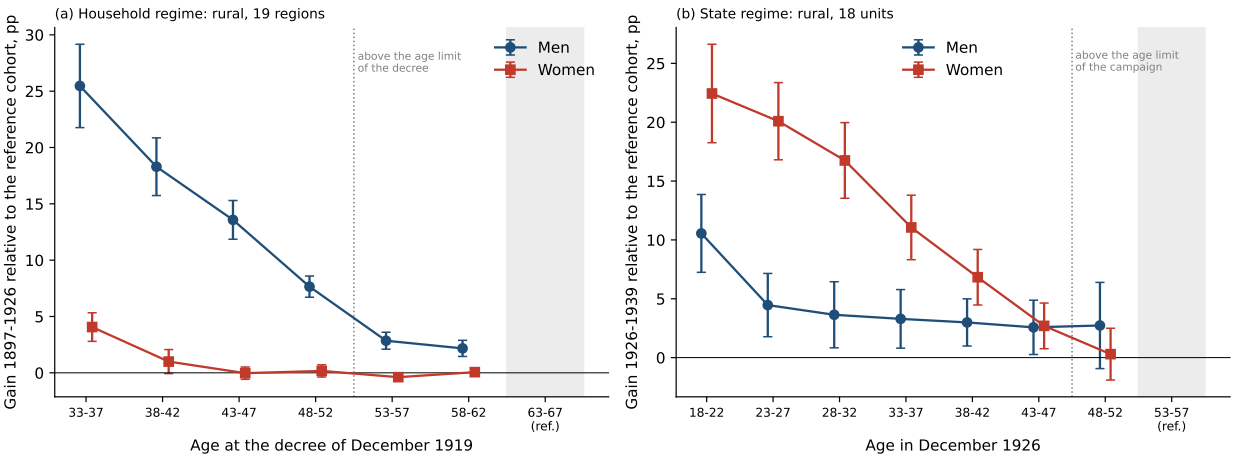
[6]Liability to service ended at the age of 43. A man aged 43 to 47 at the decree had been between 37 and 42 in August 1914, and a man aged 53 or more had been at least 47.

Table A4: Household regime: literacy gain 1897 to 1926 by age at the decree and sex, rural population

Age at the decree of 1919	Men, β_b	Women, $\beta_b + \delta_b$	Women − men, δ_b	Cell means, men / women
33-37	25.5 (1.8)	4.1 (0.6)	−21.4 (2.0)	20.9 / 2.0
38-42	18.3 (1.2)	1.0 (0.5)	−17.3 (1.4)	14.5 / −0.5
43-47	13.6 (0.8)	−0.0 (0.3)	−13.6 (0.9)	9.3 / −1.6
48-52	7.6 (0.4)	0.2 (0.3)	−7.5 (0.5)	4.0 / −1.7
53-57	2.8 (0.4)	−0.4 (0.1)	−3.2 (0.3)	−0.6 / −2.1
58-62	2.2 (0.3)	0.1 (0.1)	−2.1 (0.3)	−1.2 / −1.7
63-67 (reference)	0	1.2 (0.4)	1.2 (0.4)	−3.3 / −1.5
Aged 33-47 against 53-67, men			18.3 (1.4)	
Aged 33-47 against 53-67, women			2.0 (0.4)	
Triple difference (aged 33-47 × female)			−16.3 (1.5); excluding Muslim regions −17.3 (1.6)	
Observations; R^2			265; 0.93	

Note: Equation (A4). Percentage points; standard errors clustered by region in parentheses. Five-year age bands of 1897 linked to the same bands thirty years later in 1926 (Vol. 9, Table XV), rural population of 19 regions. Region fixed effects; weights are 1897 cohort populations. The cohort aged 63-67 at the decree is the reference. In its row the entry for women is the coefficient γ on the female indicator, the difference between women and men within the reference cohort. The cohorts aged 53 and over were above the age limit of the decree. The pooled rows compare the cohorts aged 33-47 with those aged 53-67. Cell means are unweighted. Robustness in Table A9.

Figure A6: Cohort-profile estimates of the two regimes



Note: Coefficients relative to the reference cohort with 95 per cent confidence intervals. Panel (a): equation (A4), rural population of 19 regions, reference cohort aged 63 to 67 at the decree. Panel (b): the same specification for 1926 to 1939 in 18 units, reference cohort aged 53 to 57 in 1926. The dotted line separates the cohorts above the age limit.

women and men is 2 to 3 points, against 14 to 21 in the cohorts that the decree covered. A wild cluster bootstrap over the 19 regions gives a p-value below 0.001 for the pooled triple difference.

The cohorts that gained are also the cohorts that fought in the wars of 1914 to 1921, and their survival ratios are lower for men (0.51 for the cohort born between 1878 and 1887, rural RSFSR) than for women (0.55). Differential mortality by literacy could raise the recorded literacy of survivors without any learning. A bound shows that it cannot account for the result. The excess attrition of men relative to women is 7.3 per cent of the 1897 male cohort. If every one of these excess losses had been illiterate, male literacy would have risen by 3.4 points, against an observed gain of 17.6. The cohort above the age limit, whose survival ratio is lower still (0.46), shows the selection actually realised at higher mortality, 0.4 points. The women of the cohorts that gained lost nearly as large a share of their number as the men, and their literacy did not rise, as it would have done had survival depended on literacy to an extent that mattered. Age misreporting is a second concern, since illiterate respondents and women report round ages more often (A'Hearn et al., 2009). The Whipple index at ages 23 to 62 in 1926 is 164 for illiterate men and 181 for illiterate women against 128 and 137 for the literate. The cohorts of 1926 that are linked to 1939 are centred on the ages ending in 0 and 5, so that heaping in 1926 stays within the cohort. The five-year groups of 1897 and of 1939 begin at those ages. Heaping moves into each group some of the illiterate a year or two younger and out of it some of those just below the next round age, the two movements largely offset each other, and both are present in the comparison cohorts.

E The State Regime, 1926 to 1939

E.1 Adult literacy gains

Table A5 reports the estimates for 1926 to 1939 and Figure A6(b) plots them. Relative to the cohort aged 53 to 57 in 1926, which had passed the age limit when the campaign began, rural women aged 18 to 22 gained 22 points of literacy, women aged 23 to 27 gained 20, women aged 28 to 32 gained 17, and the three cohorts that follow gained 11, 7 and 3. Men aged 18 to 22 gained 11 points and the men of every older cohort between 2.6 and 4.5. The difference between women and men is 12 to 16 points in the three cohorts aged 18 to 32, 8 and 4 points in the next two, and nothing at ages 43 to 47. In the cohorts above the age limit it is -2.4 (standard error 2.2) and 0.1 (2.0). The pooled triple difference, the cohorts aged 18 to 32 against the cohorts above the age limit and women against men, is 14.5 points (2.1), and against the cohorts aged 33 to 47 it is 9.2 (1.1).

Within cohorts the gender gap fell from 33 to 21 points for those aged 18 to 22 in 1926,

from 45 to 30 for those aged 23 to 27, and from 51 to 37 for those aged 28 to 32. For those aged 43 and over it did not move (Figure A9). The women of these cohorts had grown up under the household regime. Those aged 23 to 32 in 1926 had been 16 to 25 at the decree, 31 to 39 per cent of them could read in 1926, and those who learned afterwards did so in their late twenties and their thirties.

Table A5: State regime: literacy gain 1926 to 1939 by age in 1926 and sex, rural population

Age in 1926	Men, β_b	Women, $\beta_b + \delta_b$	Women − men, δ_b	Cell means, men / women	Gender gap, 1926 to 1939
18-22	10.6 (1.6)	22.4 (2.0)	11.9 (2.1)	14.5 / 26.6	33 to 21
23-27	4.5 (1.3)	20.1 (1.5)	15.6 (2.2)	8.0 / 23.8	45 to 30
28-32	3.6 (1.3)	16.8 (1.5)	13.1 (2.2)	7.7 / 21.0	51 to 37
33-37	3.3 (1.2)	11.1 (1.3)	7.8 (2.1)	8.2 / 16.1	51 to 43
38-42	3.0 (0.9)	6.8 (1.1)	3.8 (1.9)	8.0 / 11.9	52 to 48
43-47	2.6 (1.1)	2.7 (0.9)	0.1 (1.5)	6.4 / 6.5	51 to 51
Above the age limit of the campaign					
48-52	2.7 (1.7)	0.3 (1.0)	−2.4 (2.2)	7.9 / 5.4	41 to 43
53-57 (reference)	0	0.1 (2.0)	0.1 (2.0)	4.9 / 5.0	36 to 36
Triple difference: aged 18-32 × female, against 48-57	14.5 (2.1); leave-one-out 13.3 to 15.8; permutation p <0.001; wild bootstrap p 0.001				
Aged 33-47 × female, against 48-57	5.3 (1.6)				
Aged 18-32 × female, against 33-47	9.2 (1.1)				
Linear in age, cohorts aged 18-47	men −0.31 (0.06) per year; women −0.79 (0.06)				
Observations; units; R^2	218; 18; 0.91				

Note: Equation (A4) applied to 1926 and 1939. Percentage points; standard errors clustered by unit in parentheses. Cohorts of 1926 built from validated single years of age and linked to the five-year groups of 1939 twelve years older; rural population of 18 units. Unit fixed effects; weights are 1926 cohort populations. The cohort aged 53-57 is the reference. In its row the entry for women is the coefficient on the female indicator, the difference between women and men within the reference cohort. Cell means are weighted by population. The gender gap is male minus female literacy of the same cohort. Robustness in Table A10.

E.2 Who was taught

The share of the illiterate who were taught gives the comparison of the two regimes in its simplest form (Table A6 and Figure 2). Of the rural women aged 18 to 32 who were illiterate in 1926, 41 per cent could read in 1939, and of the men 60 per cent. Under the household regime the shares had been 0 and 30 per cent. In the cohorts above the age limit the shares under the state regime are 6 and 12 per cent, so that relative to them the state regime taught 37 per cent of the illiterate women (standard error 2.1) and 50 per cent of the illiterate men (2.5).

Two conclusions follow. The first is that the state regime did not favour women over men. An illiterate man was still more likely to be taught than an illiterate woman, by 10 to 21 points at every age below the limit. Relative to the cohorts above the limit the difference between women and men in the share taught is −12.5 points (2.7), and −10.6 (2.8) when each cohort is given its own effect. The larger gains of women in percentage points come from the composition of the illiterate. Women were 80 per cent of the illiterate of these cohorts in 1926 and 73 per cent of those who learned. The second is that the difference

Table A6: The illiterate who became literate, by regime, cohort and sex, rural population

	Men			Women			Women − men,
Cohort	Literate before	Literate after	Taught	Literate before	Literate after	Taught	taught
A. Household regime, 1897 to 1926, by age at the decree of 1919							
33-37	44	66	40.0 (3.5)	15	17	2.8 (1.0)	−37.2 (3.0)
38-42	43	58	26.9 (2.5)	14	13	−0.8 (1.0)	−27.8 (2.1)
43-47	38	49	17.4 (1.8)	11	10	−2.0 (0.7)	−19.3 (1.5)
48-52	38	42	7.7 (1.3)	9	8	−1.7 (0.7)	−9.4 (0.7)
53-57	36	36	0.0 (1.2)	8	6	−2.3 (0.5)	−2.3 (0.9)
58-62	33	32	−1.0 (1.2)	7	5	−1.8 (0.5)	−0.8 (0.8)
63-67	29	26	−4.0 (1.0)	6	4	−1.8 (0.5)	2.2 (0.7)
Aged 33-47 against 53-67			31.3 (3.0)			2.2 (0.5)	−29.1 (2.8)
B. State regime, 1926 to 1939, by age in 1926							
18-22	80	94	71.6 (1.3)	47	74	50.2 (2.3)	−21.4 (2.2)
23-27	85	93	52.6 (2.7)	39	63	39.2 (2.6)	−13.4 (2.5)
28-32	82	89	42.0 (2.3)	31	52	30.5 (2.4)	−11.6 (2.0)
33-37	76	84	33.4 (2.6)	25	41	21.4 (2.0)	−12.0 (2.7)
38-42	68	76	25.3 (2.2)	17	29	14.3 (1.9)	−11.0 (3.2)
43-47	63	69	17.2 (2.0)	12	18	7.4 (1.8)	−9.8 (2.8)
48-52	49	56	15.3 (3.1)	9	14	5.9 (2.0)	−9.4 (3.7)
53-57	43	48	8.7 (1.6)	7	12	5.4 (2.1)	−3.3 (2.5)
Aged 18-32 against 48-57			49.6 (2.5)			37.2 (2.1)	−12.5 (2.7)

Note: Literate before and after are the literacy rates of the cohort at the earlier and the later census, in per cent. Taught is the change in literacy divided by the share illiterate at the earlier census, equation (A5), in per cent, weighted by the number of illiterate at the earlier census, with standard errors clustered by region (panel A) or unit (panel B) in parentheses. The last row of each panel is the difference between the cohorts the regime could reach and the cohorts above the age limit, with region or unit fixed effects. Samples as in Tables A4 and A5.

between the regimes lies in the appearance of women among the taught. The ratio of the female share to the male share rose from 0.01 to 0.68.

The two regimes can also be compared on cohorts that were young adults when each period began. Under the household regime the cohorts aged 38 to 47 at the decree were 15 to 24 years old in 1897. Of their illiterate men 27 and 17 per cent were taught in the thirty years that followed, and of their illiterate women none. Under the state regime the cohorts aged 18 to 32 in 1926 were of similar ages at the start of the period. Of their illiterate men 60 per cent were taught in twelve years and of their illiterate women 41. The state regime therefore raised the rate at which illiterate men were taught as well, and it brought women from nothing to two thirds of that rate.

Equation (A3) gives the two shares a simple reading. If the household component of women is zero, the share of illiterate women who were taught measures the reach of the state among women, 0.41. If the reach were the same among men, their share would exceed it by $(1 - e)H_m$, where H_m is the share of men who would have learned without the campaign, at home, in the army or at work. With the share of men of the same ages taught under the household regime, 0.18, in the place of H_m, the predicted share of men is 0.52, and the observed share is 0.60. The routes that had been open to men before the campaign

thus account for 11 of the 19 points by which the share of men exceeds that of women. The remainder indicates that the instruments of the state reached men somewhat more than women, as condition (A2) allows and as the separate instruction of men about to be conscripted would produce, or that the earlier routes had widened by the 1930s. The lengths of the two periods differ, and the division should not be pressed.

Hypothesis H2 concerns age. Under the household regime the share of illiterate men who were taught fell from 40 per cent at ages 33 to 37 to nothing above the age limit, and the share of women was nil at every age. Under the state regime the shares of both sexes fall with age, by 2.21 points per year for men and 1.77 for women, and reach the level of the cohorts above the age limit between the ages of 43 and 52. The excess of men over women, as a share of the men whom a reach equal to that of women would leave out, is 43 per cent at ages 18 to 22, 22 at 23 to 27, 17 at 28 to 32 and 11 at 43 to 47. It declines with age as the hypothesis requires and as the share of men taught under the household regime had done, which was 27 per cent among those aged 15 to 19 at the start of that period, 8 among those aged 25 to 29 and nothing above 30. The decline of the share of women, which is the state's reach, is gradual, and the age limit is not visible in it as a discontinuity. A cost of teaching that rises with age produces this, and a rule applied by age alone would not.

E.3 Robustness

The sample of the state regime has 18 units, and several checks speak to the concern that so few units drive the result. The leave-one-unit-out estimates of the pooled triple difference range from 13.3 to 15.8 (Figure A10). The distribution of the coefficient under random reassignment of the sexes within unit and cohort has 95 per cent of its mass between -5.8 and 5.4, against an observed 14.5. A restricted wild cluster bootstrap over the shared 1939 source territories gives a Monte Carlo p-value of 0.001. And the analysis replicates at the level of the 17 Gosplan regions, which cover the entire RSFSR, using Table XV for 1926 and the 1939 oblasts aggregated to regions (Figure A11). At that level the pooled triple difference is 10.0 (1.6), or 8.5 (2.0) in the 12 regions whose 1926 and 1939 borders coincide, and the gender gap within cohorts falls from 31 to 23 points for those aged 20 to 24 in 1926 and from 39 to 30 for those aged 25 to 29. In the rural RSFSR as a whole 32 per cent of the women aged 20 to 34 who were illiterate in 1926 could read in 1939, and 49 per cent of the men, and women were 79 per cent of the illiterate. The shares are lower than in the sample of units, in which the cohorts are two years younger and exactly matched, and their ratio, 0.64, is close to the 0.68 of the sample.

The borders of units changed between the censuses, and the sizes of cohorts changed with them. Adding the survival ratio as a control leaves the triple difference at 13.6 (2.1).

Restricting the sample to the cells whose cohorts changed in size by no more than mortality and ordinary migration would imply, with survival ratios between 0.6 and 1.15, gives 10.4 (1.9) on 12 units. Excluding the Ural *okrugi* gives 14.6, excluding the Tatar and Bashkir republics, where Latinisation could have reduced the recorded Arabic-script literacy, gives 16.3, excluding the six low-quality units 14.8, the total population 13.3 and the cells validated at the first test alone 14.1 (Table A10). The triple difference in the share taught, with cohort effects, is between -17.1 and -8.5 across the same variations, and the share of illiterate women who were taught between 39 and 42 per cent.

The gains could be an artefact of an inflated 1939 count, since the campaign was a political target and the census that followed the suppressed census of 1937 was taken under pressure. Four features of the results speak against this reading. The definition of literacy did not change between the censuses. An enumerator who recorded the illiterate as literate at every age would not have produced a share of the illiterate recorded as literate that falls from 50 per cent of women aged 18 to 22 to 5 per cent of women aged 53 to 57, and the oldest cohorts bound what a uniform inflation could have added. The overcount of the population, which Zhiromskaya (2001) puts at 1.7 per cent, would raise a literacy rate by less than two points even if every person added had been entered as literate.

The census of 1959 is an additional consistency check, not a bound on the error of the census of 1939. Subsequent learning, loss of skills, selection and territorial changes can offset a reporting error. High literacy in 1959 therefore supports persistence among surviving cohorts without identifying when all of them learned.

The pooled population ratio of the original unit sample cannot bound mortality or out-migration because the territories differ, and some 1939 populations are reused. Accounting bounds require common territories and explicit assumptions about gross inflows. Sensitivity to these assumptions is reported separately.

Table A7: The cohorts of 1926 in the censuses of 1939 and 1959, RSFSR

	Men			Women		
Age in 1926	1926	1939	1959	1926	1939	1959
A. Rural population						
20-24	83.0	92.9	94.3	50.5	69.8	70.7
25-29	82.2	90.9	91.5	39.8	59.4	57.8
30-34	79.5	87.3	87.8	33.8	48.1	46.5
20-34	81.9	90.9	91.7	42.6	60.7	60.1
Share of the illiterate of 1926 literate, 20-34		50	54		32	31
B. Whole population						
20-24	86.0	94.8	95.8	58.0	77.3	76.5
25-29	85.2	93.2	93.7	49.1	68.8	66.0
30-34	83.1	90.5	90.6	43.5	58.9	56.1
20-34	85.0	93.2	93.8	51.2	69.7	67.6
Share of the illiterate of 1926 literate, 20-34		55	59		38	34

Note: Literate persons per hundred. The 1926 figures are those of Table XV of Volume 9 for the 17 regions that formed the RSFSR of 1939; the 1939 and 1959 figures are sums over the RSFSR. A five-year group of 1926 is twelve years older in January 1939 and thirty-two years older in January 1959, and is matched in each census to three fifths of one five-year group and two fifths of the next. In 1959 the literate are those with any schooling and those without schooling who can read. The last row of each panel gives the change since 1926 as a share of those illiterate in 1926.

Table A8: The five-year groups of 1926 in the census of 1939, RSFSR

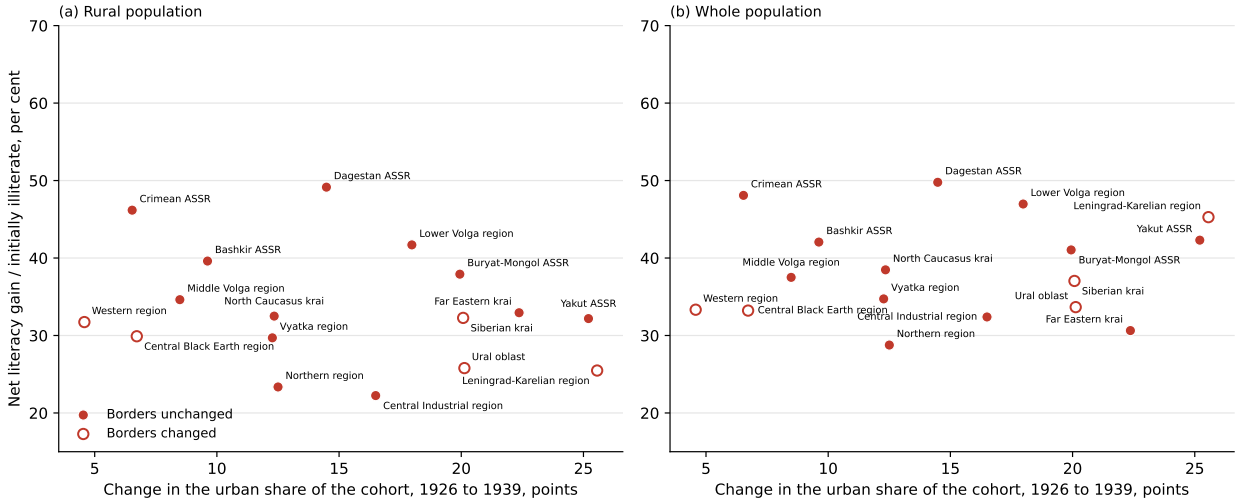
	Rural population						Whole population					
	Women			Men			Women			Men		
Age in 1926	1926	1939	Taught	1926	1939	Taught	1926	1939	Taught	1926	1939	Taught
20-24	50.5	69.8	39	83.0	92.9	58	58.0	77.3	46	86.0	94.8	63
25-29	39.8	59.4	33	82.2	90.9	49	49.1	68.8	39	85.2	93.2	54
30-34	33.8	48.1	22	79.5	87.3	38	43.5	58.9	27	83.1	90.5	43
35-39	26.2	36.4	14	75.1	82.3	29	35.1	47.5	19	79.1	86.4	35
40-44	18.1	25.0	8	68.7	75.5	22	26.2	35.8	13	73.4	80.7	27
45-49	13.7	17.7	5	60.9	65.3	11	21.2	26.8	7	66.1	71.6	16
50-54	10.2	13.4	3	52.5	56.3	8	17.0	21.7	6	58.2	62.5	10

Note: Literate persons per hundred in 1926 (Table XV of Volume 9, the 17 regions that formed the RSFSR of 1939) and in 1939 (sums over the RSFSR, each five-year group of 1926 being matched to three fifths of one five-year group of 1939 and two fifths of the next). Taught is the change as a share of those illiterate in 1926, in per cent.

F Regions and the Move to the Towns

Figure A7 shows the regional test of Section 5. Each point is one of the 17 regions that made up the RSFSR of 1939, and the horizontal axis is the change in the share of the cohort aged 20 to 34 in 1926 that lived in towns, a measure of how far the opportunities of the new economy reached the region and of how many of its young people left the village.

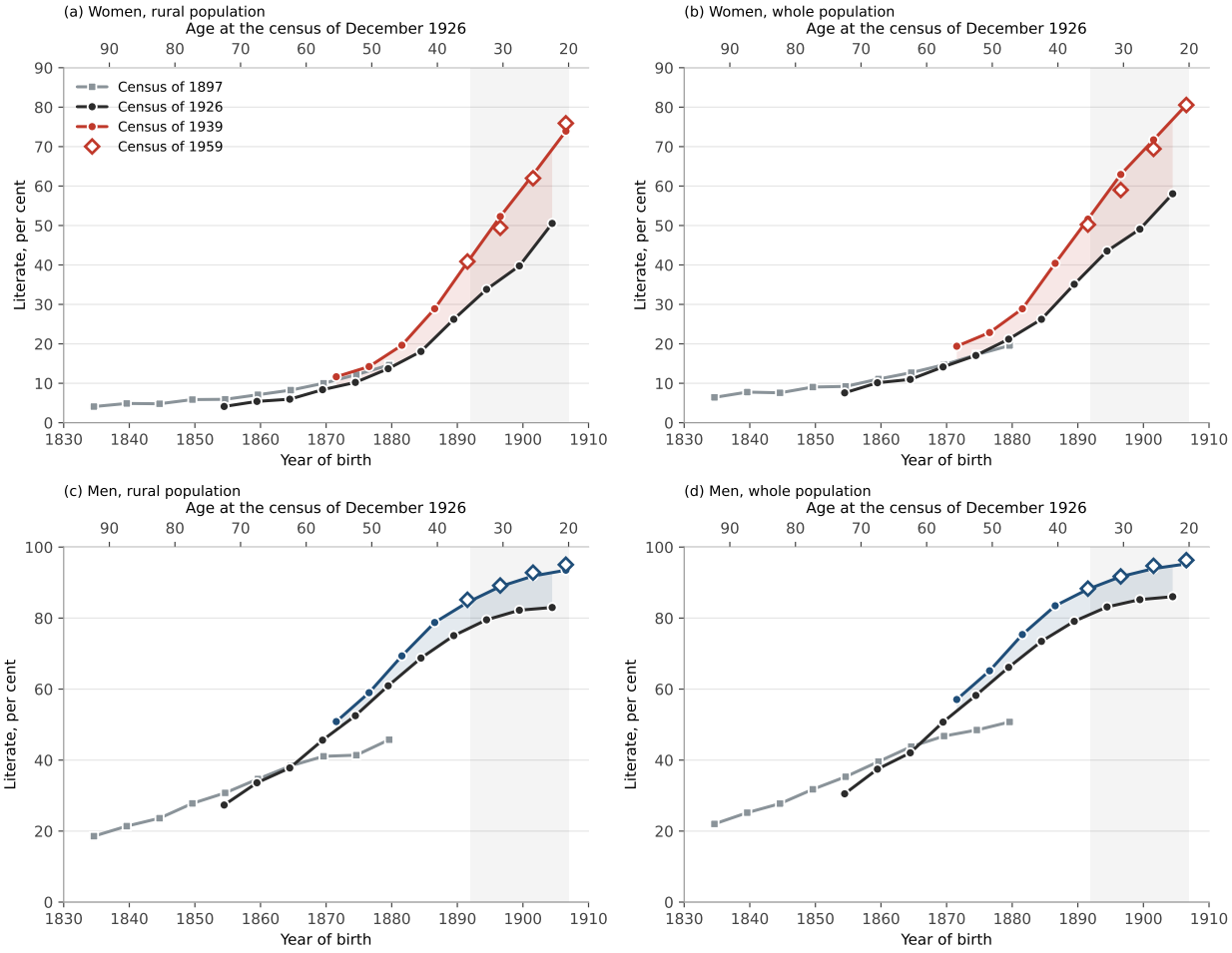
Figure A7: The share of illiterate women who learned and the move of their cohort to the towns, by region



Note: Cohort aged 20 to 34 in 1926, women. The vertical axis is the change in literacy between 1926 and 1939 divided by the share illiterate in 1926. The horizontal axis is the change in the urban share of the same cohort, computed as one minus the ratio of its rural to its total number in each census. Filled points mark the regions whose borders did not change. Panel (a): rural population. Panel (b): whole population.

G Additional Tables and Figures

Figure A8: Literacy by year of birth as recorded by four censuses, by sex and population



Note: As Figure 1, which is repeated in panel (a), for the whole population of the RSFSR and for men. Each point is a five-year age group of the census named, placed at its mean year of birth; the upper axis gives the age of the cohort in December 1926 and the grey band marks the cohort aged 20 to 34. The censuses of 1939 and 1959 are drawn in red for women and in blue for men. Among men the census of 1926 lies above that of 1897 for the cohorts born after 1865, who learned as adults before the campaign.

Table A9: Household regime: robustness of the estimates by cohort

Specification	Men 33-42	Men 43-52	× female 33-42	× female 43-52	n
Baseline (rural, weighted; ten-year cohorts, reference 53-62)	19.2 (1.7)	8.0 (0.7)	−16.5 (1.9)	−7.8 (0.7)	113
Unweighted	18.5 (1.6)	7.5 (0.7)	−15.8 (1.5)	−7.3 (0.7)	113
With survival ratio	19.5 (1.9)	8.4 (1.0)	−16.7 (2.0)	−8.0 (0.8)	113
Excluding Muslim regions	20.3 (1.9)	8.5 (0.7)	−17.4 (2.1)	−8.3 (0.8)	84
Survival ratio in $[0.4, 0.8]$	19.0 (1.8)	8.1 (0.9)	−16.2 (2.1)	−8.0 (1.0)	90
Total population	20.0 (1.7)	8.1 (0.6)	−15.5 (2.0)	−7.3 (0.7)	113
Urban population	3.7 (2.0)	5.9 (0.9)	−6.6 (0.8)	−5.5 (0.9)	114
Five-year bins (Table A4), δ_b for 33-37 / 38-42 / 43-47	−21.4 (2.0) / −17.3 (1.4) / −13.6 (0.9)				
Five-year bins, unweighted	−20.7 (1.7) / −16.8 (1.3) / −12.8 (1.2)				
Five-year bins, with survival ratio	−21.5 (2.0) / −17.6 (1.3) / −14.0 (0.9)				
Five-year bins, total population	−20.1 (2.2) / −16.4 (1.6) / −12.4 (0.9)				
Triple difference without the cohort aged 33-37	−13.9 (1.2); men aged 38-47 against 53-67: 14.5 (1.1)				

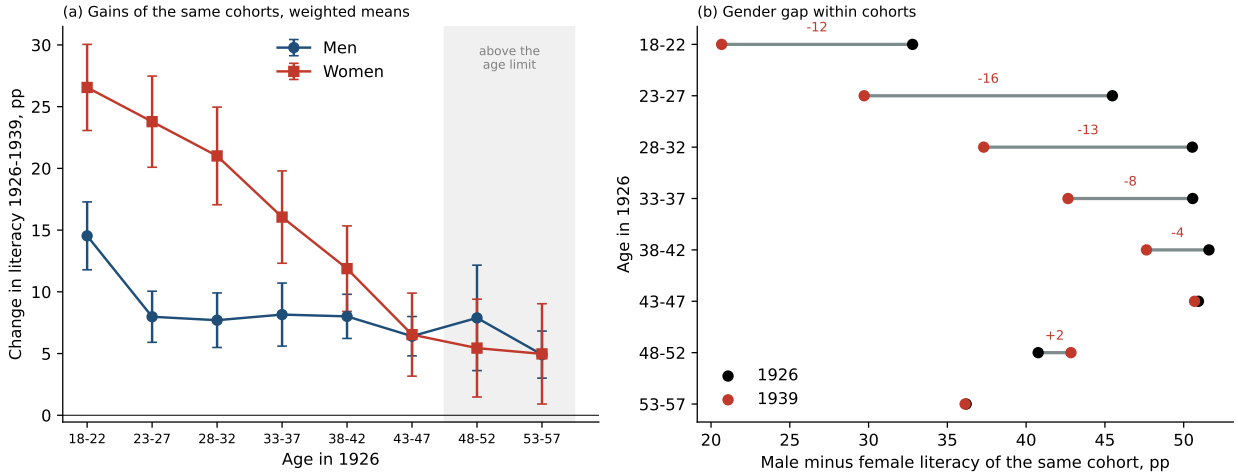
Note: Percentage points; standard errors clustered by region in parentheses. Upper panel: ten-year cohorts aged 33-42 and 43-52 at the decree against the cohort aged 53-62; region fixed effects. Lower panel: equation (A4) with five-year bins.

Table A10: State regime: robustness of the difference between women and men by cohort

	Female excess gain δ_b by age in 1926							Triple	Cells;
Specification	18-22	23-27	28-32	33-37	38-42	43-47	48-52	difference	units
Baseline	11.9 (2.1)	15.6 (2.2)	13.1 (2.2)	7.8 (2.1)	3.8 (1.9)	0.1 (1.5)	−2.4 (2.2)	14.5 (2.1)	218; 18
Including flagged units	10.6 (2.3)	14.0 (2.6)	11.7 (2.4)	6.5 (2.3)	3.1 (1.9)	−0.4 (1.5)	−2.4 (1.9)	13.1 (2.4)	250; 20
With survival ratio	11.4 (2.2)	15.0 (2.3)	12.7 (2.3)	7.6 (2.2)	3.6 (1.9)	1.1 (1.4)	−2.4 (2.1)	13.6 (2.1)	218; 18
Survival ratio in $[0.6, 1.15]$	8.0 (2.4)	11.4 (2.2)	8.8 (2.1)	3.6 (1.8)	−0.3 (1.3)	−2.2 (2.0)	−2.0 (1.5)	10.4 (1.9)	124; 12
Unweighted	11.8 (1.9)	14.9 (1.9)	12.0 (1.9)	6.0 (1.8)	1.3 (1.6)	−1.5 (1.4)	−3.0 (2.2)	14.2 (2.1)	218; 18
Russian units only	13.4 (1.8)	17.1 (2.1)	14.5 (2.4)	9.8 (2.5)	5.6 (2.4)	1.6 (1.4)	−0.3 (1.7)	14.9 (2.0)	138; 12
Excluding Ural okrugi	11.9 (2.2)	15.7 (2.3)	13.2 (2.3)	8.0 (2.3)	4.1 (2.0)	0.3 (1.6)	−2.5 (2.3)	14.6 (2.2)	178; 15
Excluding Tatar and Bashkir republics	14.1 (1.6)	17.5 (1.8)	14.8 (2.1)	9.6 (2.0)	4.7 (2.0)	1.1 (1.2)	−1.9 (2.2)	16.3 (1.9)	194; 16
Excluding low-quality units	12.7 (2.1)	16.5 (2.2)	14.0 (2.3)	8.0 (2.2)	4.0 (1.9)	0.3 (1.6)	−1.4 (2.5)	14.8 (2.5)	192; 15
Cells validated by the first rule only	12.1 (2.1)	15.6 (2.2)	13.1 (2.2)	7.8 (2.1)	3.8 (1.9)	0.1 (1.5)	−1.5 (2.2)	14.1 (2.3)	214; 17
Units with all eight cohorts	13.8 (3.4)	17.9 (3.0)	15.4 (3.2)	8.9 (2.9)	5.4 (2.2)	2.0 (0.9)	0.3 (0.4)	15.4 (3.2)	112; 7
Total population	12.1 (1.3)	15.3 (1.3)	12.1 (2.4)	7.5 (1.3)	5.1 (1.5)	0.8 (1.0)	−0.1 (0.4)	13.3 (1.4)	352; 25

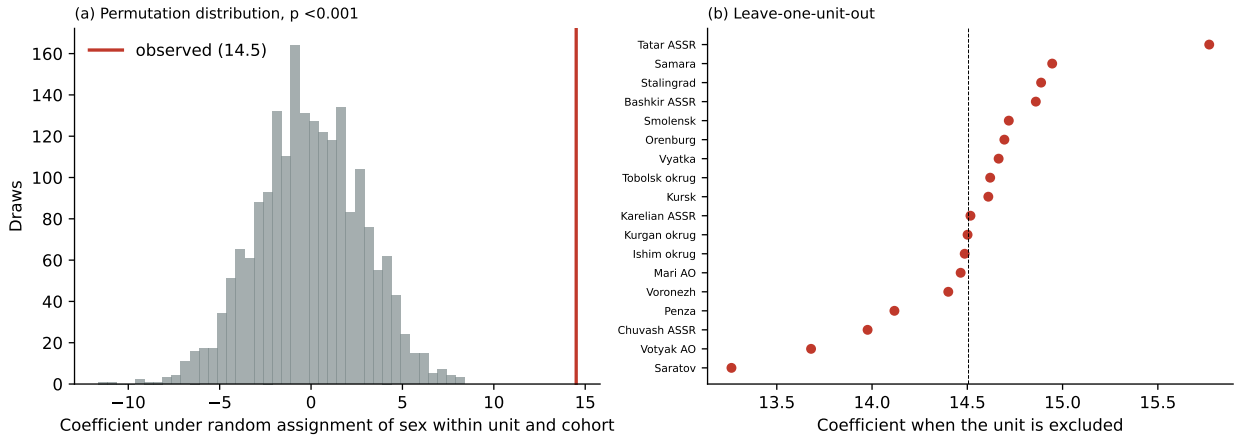
Note: Coefficients δ_b of equation (A4) applied to 1926 and 1939, relative to the cohort aged 53-57, and the pooled triple difference of the cohorts aged 18-32 against those aged 48-57. Unit fixed effects; standard errors clustered by unit.

Figure A9: The state regime: gains of the same cohorts and the gender gap within cohorts



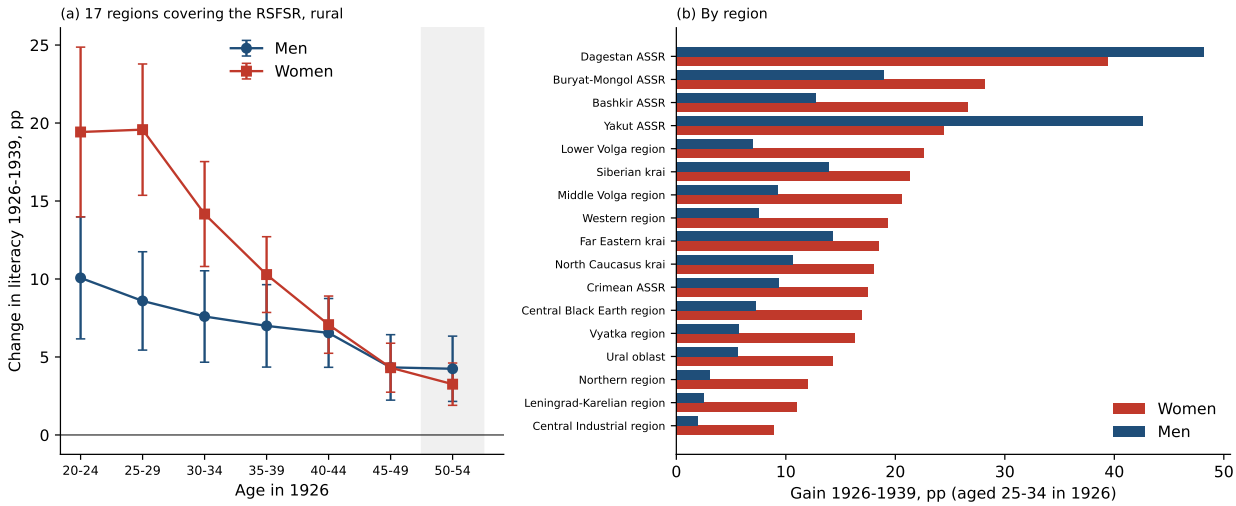
Note: Panel (a): population-weighted mean gains with 95 per cent confidence intervals, rural population of 18 units. Panel (b): male minus female literacy of the same cohort in 1926 and in 1939.

Figure A10: Inference for the triple difference of the state regime



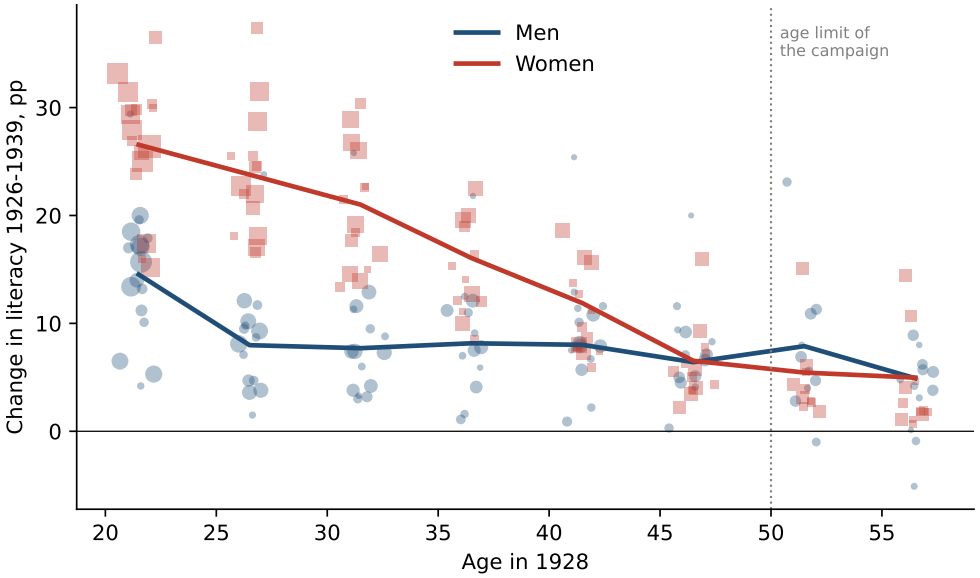
Note: Panel (a): distribution of the pooled triple-difference coefficient under 2,000 random reassignments of the sexes within unit and cohort. Panel (b): the coefficient when each unit in turn is excluded.

Figure A11: Replication of the state regime at the level of Gosplan regions



Note: 1926 from Vol. 9, Table XV; 1939 oblasts aggregated to the 17 regions. Because the five-year bands of 1926 shift by twelve years, each band is matched to a mixture of the two overlapping bands of 1939 in the proportion three to two. The shaded cohort was above the age limit.

Figure A12: Gains under the state regime by unit and cohort



Note: Cells by unit and cohort, rural population, sized by 1926 population; lines join population-weighted means.

H New regional follow-up and institutional evidence

Regional cohort literacy after 1939

Regional tabulations of the 1959 census allow a more demanding follow-up than the national comparison alone. I compare the cohorts aged 30–34, 35–39 and 40–44 in January 1939 with those aged 50–54, 55–59 and 60–64 in January 1959, holding their age composition fixed at the 1939 census. These age bands can be matched without interpolation. They overlap with, but do not exactly reproduce, the paper's group aged 20–34 in December 1926. The five-republic diagnostic covers the Tatar, Bashkir, Chuvash, Mari and Udmurt republics. Karelia is excluded because of territorial comparability concerns. The correspondence follows the republics' territorial cores; a complete concordance of small territorial transfers has not been established.

Across these five republics, female literacy in the selected cohorts increases by 2.07 percentage points for the total population and by 3.78 points for the rural population, but changes by -8.10 points in towns. The corresponding changes for men are 3.00, 3.49 and -0.32 points. An exploratory regression of the cohort-level change on a female indicator, republic and age-group fixed effects, weighted by population in 1939, gives a female-minus-male change of -7.66 points in towns. The CR2/Satterthwaite 95 per cent interval is [-12.48, -2.83]; the restricted wild cluster bootstrap, enumerating all 32 Rademacher sign vectors for the five republics, gives $p = 0.0625$. The two procedures give different answers at the conventional five per cent threshold, so both are reported.

These are changes in the recorded literacy of populations occupying successive age groups, not transitions observed for linked individuals. For example, the Tatar republic's female urban cohort population is 120.9 per cent of its 1939 size by 1959. Migration, settlement reclassification, territorial change or enumeration error must therefore enter the comparison. Wartime mortality, selective survival, subsequent learning and possible loss of skills add further components. Consequently, similarity between the two national literacy rates cannot by itself bound misreporting in 1939 or establish uniform regional persistence. The post-1939 gender-gap estimates are descriptive and do not identify the causal effect of the literacy campaign.

H.1 Institutions in 1924 and the age target of the 1931 plan

Table XVII of the Central Statistical Administration's survey of educational institutions on 1 January 1924 reports literacy schools and points, teachers and pupils by sex, separately for rural areas. I transcribe 30 rows for 15 territories from printed pages 462–467 of *Narodnoe*

obrazovanie v SSSR, vol. XXVIII, issue 1 (1926). General adult schools in adjacent columns are excluded. Fourteen territories can be linked to validated male and female cells aged 18–22 in 1926. The Bashkir republic lacks this validated cohort in the estimation sample. No duplicated oblast exposure is assigned to the individual Ural districts.

The counts describe institutions and pupils at a date, rather than unique graduates or exposure of a particular age cohort. Coverage is heterogeneous: only five of 23 rural Mari institutions were surveyed, compared with all 81 in the Tatar republic. The total Bashkir teacher count exceeds the sum of the two reported sexes by three; the transcription preserves this unresolved discrepancy. Territorial transfers between 1924, 1926 and 1939 have not been fully harmonised.

Exploratory models relate the subsequent female-minus-male literacy gain to female shares of surveyed pupils or teachers, controlling for the initial literacy gap. Other models use log counts of institutions or teachers, with cohort population as a territory-size control. This population is not treated as the full eligible population or used as an enrolment-rate denominator. Across 24 combinations of exposures, weights and coverage restrictions, every CR2/Satterthwaite 95 per cent interval includes zero. Results do not establish a robust institutional mechanism; small samples, incomplete observation and endogenous allocation also prevent interpreting them as evidence of no effect.

The 1931 national economic plan specifies ages 16–50 in towns and 16–40 in rural areas, with some instruction above 40 in collectivised areas (printed pages 67–68). Planned contingents are not observed graduates. As a sensitivity check, dropping the cohort aged 33–37 in 1926, whose ages straddle 40 in 1931, changes the main female contrast from 20.354 to 20.312 percentage points. The younger group remains aged 18–32 in both models; ages 33–37 belong to the intermediate group. After their omission, the female contrast has a CR2 interval of [16.606,24.019]. The triple difference changes from 14.379 to 14.377. This checks the inclusion of one intermediate cohort; it does not establish an untreated older comparison group or remove migration and reporting bias.

Machine-readable transcriptions, source page images, all specifications and fixed seeds are supplied in `supplementary/likbez_1924_analysis/`. The full source is available at https://istmat.org/files/uploads/22102/obrazovanie_1921-25_vyp.1_1924_god.pdf; the 1931 plan at https://istmat.org/files/uploads/52124/plan_sssr_na_1931_god.pdf.